
Just Ramp-Up: Debiasing Regression-based Estimator for A/B Tests under Network Interference

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 Network interference complicates A/B testing on online platforms, such as social
2 networks and marketplaces, where causal methods based on single experiments
3 often suffer from significant bias due to complex interference patterns. This paper
4 demonstrates the statistical benefits of merging data from multiple experiments
5 with varying treatment proportions. Sequential experimentation with increasing
6 traffic, or ramp-up, is widely used in tech companies for risk management and cost
7 control. Beyond operational benefits, we show that regression-based estimators
8 trained on merged data achieve substantial bias reduction, even under simple
9 randomization schemes and regression models. We focus on the global average
10 treatment effect (GATE), a key estimand in the tech industry, and consider a general
11 interference pattern that extends beyond the 1-hop setting. We present a closed-form
12 bias-variance analysis of the linear regression estimator and show that, in practical
13 settings, the bias term dominates. In addition, we characterize the substantial
14 bias reduction achieved by our proposed approach, which merges experimental
15 data collected at different ramp-up stages to improve the training of the regression
16 function. We also offer an intuitive explanation for this reduction and highlight
17 the synergy between cluster-level randomization and our approach. Furthermore,
18 we consider a refined estimator based on graph neural networks (GNN). Extensive
19 simulations across challenging scenarios confirm that our methodology significantly
20 improves the accuracy of regression-based estimators.

21 1 Introduction

22 A/B testing has become a core paradigm for data-driven decision-making on modern online platforms.
23 In many applications, the target causal estimand is the global average treatment effect (GATE),
24 which measures the difference in average outcomes between globally deploying a new feature and
25 maintaining the status quo. In complex systems such as social networks and marketplaces, however,
26 this task is complicated by network interference: a unit's outcome may depend not only on its own
27 treatment, but also on the treatments assigned to other connected units [25, 22, 13].

28 Estimating GATE under network interference is particularly challenging in practice because platforms
29 rarely expose all users to treatment during experimentation. Instead, new features are typically rolled
30 out gradually through a ramp-up process, starting from a very small treatment proportion and
31 increasing traffic only for promising variants [31, 17]. As a result, the available data are collected
32 under limited treatment proportions, whereas GATE concerns two extreme counterfactual regimes:
33 global treatment and global control. This makes GATE estimation fundamentally an extrapolation
34 problem, and interference further amplifies its difficulty [21].

35 In this paper, we show that the ramp-up process provides not only operational benefits but also a
36 valuable statistical opportunity. We propose to merge data from experiments at different ramp-up

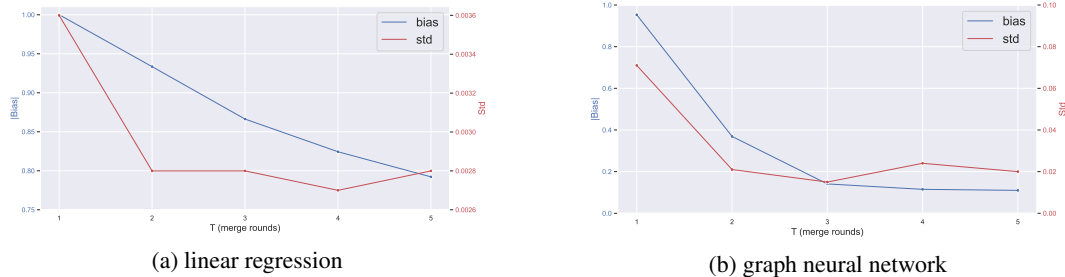


Figure 1: The trajectory of bias (left y-axis) and standard deviation (right y-axis) of the regression-based estimators trained on the **merged** T -step experimental data. Here we set the treatment proportions $(c_1, c_2, \dots, c_5) = (2\%, 5\%, 10\%, 25\%, 50\%)$. The t -th column in the figure corresponds to the case that the regression model is trained on merged data with treatment proportions $(c_{6-t}, \dots, c_5)$.

37 stages and use the pooled data to train a regression model for counterfactual prediction. The key
 38 advantage of this merging procedure is that it enriches the variation in units' treatment exposures, i.e.,
 39 summary statistics of interference on a given unit, such as the proportion of treated neighbors in its
 40 1-hop neighborhood, thereby helping the regression model extrapolate to global treatment.

41 Our focus is on regression-based estimators for static GATE. Beyond their practical advantages, such
 42 as ease of implementation and the ability to incorporate covariates [11], regression-based estimators
 43 are especially attractive here because they can accommodate substantially richer interference struc-
 44 tures than traditional estimators without a regression component. Much of the existing literature relies
 45 on neighborhood interference or related low-dimensional assumptions, which reduce the effect of the
 46 full treatment vector on a unit to the treatments within its 1-hop neighborhood [10, 7, 27, 20, 15, 5].
 47 While such assumptions offer important technical simplifications, spillover beyond the 1-hop neigh-
 48 borhood remains underexplored despite its practical relevance. To address this gap, we study a novel
 49 interference pattern, *general linear interference*, which captures multihop spillover while still permit-
 50 ting *closed-form* bias and variance analysis, in contrast to the classical approximate neighborhood
 51 interference framework [19], which puts bounded-difference control on the impact of long-distant
 52 neighbors. Our analysis also incorporates the network topology explicitly, rather than relying only on
 53 coarse graph summaries such as the maximum degree, as is common in the literature.

54 We also emphasize the role of multiple experiments. While ramp-up experiments are standard in
 55 industry, most existing methods for network interference focus on a single experiment. There are
 56 some notable exceptions. For example, [7] models the mean outcome as a one-dimensional function
 57 of the treatment proportion and extrapolates to global treatment and control, while [3] uses multiple
 58 experiments for model selection over candidate interference structures. By contrast, our contribution
 59 is to show that multiple experiments can be pooled to improve regression-based GATE estimation
 60 itself. We interpret GATE estimation as a counterfactual prediction problem between two extreme
 61 exposure regimes, and the central benefit of merging lies in improving the training of the regression
 62 function through **richer network exposure variation** among units.

63 Our theoretical analysis begins with a simple network-agnostic linear regression estimator under our
 64 novel general linear interference model. We adopt linear regression because, despite the complexity
 65 of long-range network interference, it offers valuable tractability for *closed-form* bias and variance
 66 analysis. Moreover, the resulting insights extend to the behavior of more refined regression-based
 67 estimators. We derive a closed-form bias–variance decomposition and show that, in practically
 68 relevant regimes with regular interference intensity, the estimation error is typically dominated by
 69 bias. We further characterize how merging data across ramp-up stages can substantially reduce
 70 this bias by increasing the variation in treatment proportions and, consequently, in units' treatment
 71 exposures.

72 Motivated by this finding, we then study a more expressive estimator based on graph neural networks
 73 (GNNs). Extensive simulations show that the same qualitative pattern persists: merged data markedly
 74 improve estimation accuracy, and powerful models such as GNNs benefit especially strongly from
 75 the richer exposure variation induced by merging. In this paper, we intentionally examine the
 76 performance of lightweight models, in contrast to GNN-based approaches with carefully designed
 77 architectures [29, 12]. Our key message is that, with our data-centric ramp-up augmentation, even
 78 simple regression models can achieve strong performance in GATE estimation.

Figure 1 provides an illustrative example. For both linear regression and GNN estimators, merging data from multiple ramp-up stages leads to a clear reduction in bias, while the variance remains comparatively small. The figure also highlights a key practical message: even expressive models such as GNNs perform unsatisfactorily when trained on a single experiment, but become much more effective once trained on merged datasets with distinct treatment proportions.

Our main takeaway is therefore simple: by pooling ramp-up experiments with different treatment proportions, one can train a regression model that more accurately predicts potential outcomes under global treatment and global control. Since ramp-up is already standard practice on many online platforms, this methodology requires no additional experimentation cost and can be deployed as a plug-and-play improvement for GATE estimation under network interference.

2 Basic Setting

2.1 Preliminary

We adopt the Neyman-Rubin potential outcome framework [24] and consider a finite population of n units. The treatment vector is represented as $\mathbf{z} = (z_1, z_2, \dots, z_n) \in \{0, 1\}^n$. Without assuming the Stable Unit Treatment Value Assumption (SUTVA), we define the potential outcome of unit i as $Y_i(\mathbf{z})$ to account for its possible dependency on the treatments of other units. The parameter of interest is the global average treatment effect (GATE), which is defined as:

$$\tau := \frac{1}{n} \sum_{i \in [n]} (Y_i(\mathbf{1}) - Y_i(\mathbf{0})). \quad (1)$$

Here, $[n]$ denotes the unit set $\{1, 2, \dots, n\}$, and $\mathbf{1}(\mathbf{0})$ denotes the n -dimensional all-ones (all-zeros) vector, corresponding to global treatment and global control, respectively. We take the network structure as given, defined by an undirected graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$. Here the node set $\mathcal{V}$ is the unit set $[n]$, and edges are characterized by the adjacency matrix A . In addition, we use $\deg_i$ to denote the degree of node i on the graph, and matrix $D = \text{diag}(\deg_1, \deg_2, \dots, \deg_n)$. For convenience, we assume $\deg_i > 0$ for all unit $i \in [n]$.

2.2 General Linear Interference Model

We consider the following data generation mechanism, called *general linear interference*:

$$Y(\mathbf{z}) = \beta_0 \mathbf{1} + \beta_1 \mathbf{z} + B\mathbf{z} + \epsilon, \quad (2)$$

where (β_0, β_1) and B are all unknown parameters. With these, the true GATE is given by:

$$\tau = \beta_1 + \frac{1}{n} \mathbf{1}^\top B \mathbf{1}, \quad (3)$$

For convenience, we further assume an i.i.d. mean-zero noise structure.

$$\mathbb{E}[\epsilon] = \mathbf{0}, \quad \text{Var}(\epsilon) = \sigma_e^2 I_n, \quad \mathbf{z} \perp \epsilon. \quad (4)$$

The matrix B in Equation (2) characterizes interference effects. In this paper, we impose $B_{ii} = 0$ for all $i \in [n]$ to separate the direct treatment effect from the interference effect. For simplicity, we do not model heterogeneity in the baseline effect β_0 or direct effect β_1 across units. Instead, we place the modeling flexibility on a general interference matrix B , which can capture complex and potentially long-range dependencies on the network. We provide two examples of such matrices B below.

Example 2.1 (1-hop linear potential outcome model). If we take $B = rD^{-1}A$, where r is the scale parameter, then our potential outcome model is reduced to the classic linear-in-means model [1]:

$$Y_i(\mathbf{z}) = \beta_0 + \beta_1 z_i + r \frac{\sum_{j \in \mathcal{N}(i)} z_j}{\deg_i} + \epsilon_i, \quad (5)$$

where $\mathcal{N}(i)$ denotes the 1-hop neighborhood of unit i in the graph.

Example 2.2 (Multihop linear potential outcome model). We consider

$$B = (\mathbf{1}\mathbf{1}^\top - I_n) \odot \left(\sum_{m=1}^M r_m (D^{-1}A)^m \right), \quad (6)$$

where $\odot$ denotes the Hadamard product, M is the interference range, and $r_1 \geq r_2 \geq \dots \geq r_M > 0$.

Notice that A^m characterizes the m -hop connections on the graph, where A_{ij}^m represents the number of walks of length m connecting units i and j . The term D^{-1} is included here as a normalization factor. The sequence of scale parameters, $\{r_i\}_{i=1}^M$, describes how interference decreases with increasing distance on the graph. Additionally, we exclude the diagonal elements of matrix B . This exclusion addresses the identification issue and can also be interpreted as eliminating the circular effects associated with closed walks that begin and end at the same unit. Since $D = \text{diag}(A1)$, we can summarize the examples above as $B = g(A)$ for certain function g , namely, B models the interference that is conducted through the network topology. We consider the proposed general linear model for carrying out *closed-form* bias and variance analysis while maintaining sufficient generality.

2.3 Experimental Design

In this section, we discuss our default experimental design. Specifically, we adopt unit-level *complete randomization* as the basic randomization scheme, which randomly chooses exactly d units from a population of n units to receive treatment. To formalize it, we uniformly sample the binary treatment vector $\mathbf{z}$ from $\{\mathbf{z} \in \{0, 1\}^n : \sum_{i=1}^n z_i = d\}$. In particular, $\mathbb{E}[z_i] = d/n$.

We note that the performance of unit-level complete randomization is often very similar to the independent Bernoulli randomization, which assigns treatment to each unit with probability $p = d/n$ independently. However, unit-level complete randomization ensures exactly d treated units, which brings considerable convenience in our derivation for closed-form expressions of bias and variance.

We next discuss the setting of multiple experiments. We consider a ramp-up process that includes T experiments that are subject to complete randomization with treated units $d_1 < d_2 < \dots < d_T$. For convenience, we introduce another notation $c_t = d_t/n$. For instance, $T = 5$ and $(c_t)_{t=1}^T = (2\%, 5\%, 10\%, 25\%, 50\%)$, as also illustrated in [31]. Since our core methodology is merging multiple experimental data, we will also discuss the selection of T and $(c_t)_{t=1}^T$ later in the analysis.

2.4 Regression-based Estimator

We consider a regression-based estimator based on a function $f \in \mathcal{F}$ for predicting the counterfactual $Y(\mathbf{z})$, where $f : \{0, 1\}^n \times \mathcal{A} \rightarrow \mathbb{R}^n$ and $\mathcal{A}$ is the space of adjacency matrices of simple graphs on n nodes. Given a prediction function f trained on experimental data, we define the estimator as the difference between the predicted mean outcomes under global treatment and global control:

$$\hat{\tau}(f) = \frac{1}{n} \mathbf{1}^\top (f(\mathbf{1}, A) - f(\mathbf{0}, A)). \quad (7)$$

In classical causal inference, the augmented inverse propensity weighting (AIPW; [6]) estimator is generally preferred to purely regression-based estimators. Under network interference, however, inverse propensity weights often lead to explosive variance [27, 5] and are also vulnerable to exposure misspecification. Regression-based estimators therefore occupy a distinctive and important role. In this paper, we focus on two regression model classes, linear regression $\mathcal{F}_{\text{LR}}$ and graph neural networks $\mathcal{F}_{\text{GNN}}$, and develop a data-centric approach for bias reduction in regression-based estimation.

3 Analysis on Linear Regression Estimator

In this section, we analyze the bias and variance of a graph-agnostic and deliberately misspecified linear regression model under the general linear network interference framework introduced above. As shown below, this model provides a clean illustration of how merging experimental data affects training. Moreover, many phenomena observed in this simple setting extend to more complex models, such as those in $\mathcal{F}_{\text{GNN}}$, as demonstrated in our simulation study. Overall, this linear regression model is both technically tractable and sufficiently expressive to yield generalizable insights.

Specifically, we consider the estimator $f(\mathbf{z}, A) = X(\mathbf{z})\hat{\theta}$, where $X(\mathbf{z}) = (\mathbf{1}, \mathbf{z})$ and $\hat{\theta}$ is the ordinary least squares estimator

$$\hat{\theta} = (X(\mathbf{z})^\top X(\mathbf{z}))^{-1} X(\mathbf{z})^\top Y. \quad (8)$$

We retain the notation $X(\mathbf{z})$ to emphasize its dependence on the experimental design.

Linear regression is technically convenient because it admits a closed-form estimator, unlike many regression-based methods defined via variational formulations. The main difficulty is that the design

matrix $X(\mathbf{z})$ is itself random under treatment assignment, so $(X(\mathbf{z})^\top X(\mathbf{z}))^{-1}$ contains a random determinant in the denominator, severely hindering exact analysis. However, we find that *complete randomization* offers a crucial source of tractability by fixing the number of treated units.

3.1 One-step Experiment

Throughout the section, we use $\hat{\tau}_{(T)}$ to denote the linear regression estimator based on merged T -step experiments. We now examine the case of $T = 1$ as a preparation. We specify the number of treated units in the experimental data as $d = cn$ and derive the bias and variance for $\hat{\tau}_{(1)}$.

Theorem 3.1. *The bias and variance of $\hat{\tau}_{(1)}$ under general linear interference are given by:*

$$\text{Bias}(\hat{\tau}_{(1)}) = -\frac{(\sum_{i,j} B_{ij})}{n-1}, \quad (9)$$

and

$$\begin{aligned} \text{Var}(\hat{\tau}_{(1)}) = & \left(\frac{1}{nc(1-c)} \right)^2 \left(\left(\sum_{i,j} B_{ij} \right)^2 \left(\frac{c^3(c-1)}{n} + O\left(\frac{1}{n^2}\right) \right) \right. \\ & + \left(\sum_{i,j,k} B_{ij} B_{kj} \right) \left(\frac{c(c-1)}{n} + O\left(\frac{1}{n^2}\right) \right) + \left(\sum_{i,j,l} B_{ij} B_{il} \right) \left(c^3(1-c) + O\left(\frac{1}{n}\right) \right) \\ & + \left(\sum_{i,j} B_{ij}^2 \right) \left(c^2(1-c) + O\left(\frac{1}{n}\right) \right) + \left(\sum_{i,j} B_{ij} B_{ji} \right) \left((1-c)^2 c^2 + O\left(\frac{1}{n}\right) \right) \\ & \left. - \left(\sum_{i,j} B_{ij} \right)^2 \frac{1}{(n(n-1))^2} + \frac{\sigma_e^2}{nc(1-c)} \right). \end{aligned} \quad (10)$$

Here, notations like $\sum_{i,j}$ denote summation over $i \in [n], j \in [n]$ without additional constraints. We suppress lower-order terms for readability; the complete expansion is given in Appendix C.1, and the hidden constants are indeed small. We now proceed to elaborate on the results presented above. First, we notice that the intensity of direct effect, β_1 , does not influence either bias or variance. In total, the primary source of bias is interference, which corresponds to the second term in the expression.

Although the variance expression involves several quadratic functionals of B , the main takeaway is simple: under a regular interference regime (the interference is comparable to the direct effect), all of these terms are controlled at a lower order than the squared bias. This motivates the next assumption, which compresses the relevant requirements into three interpretable conditions.

Assumption 3.2 (Regular interference). With general linear interference, we further assume that

$$\sum_{j=1}^n |B_{ij}| = O(1) \quad \forall i \in [n]. \quad (11)$$

Moreover,

$$\sum_{i,j} B_{ij} = \Theta(n) \quad \sum_j \left(\sum_i B_{ij} \right)^2 = O(n). \quad (12)$$

The three conditions serve distinct purposes: the first bounds the total spillover received by each unit, the second ensures that average interference remains non-negligible, and the third rules out a vanishing fraction of highly influential nodes from dominating the variance. Under this regime, the variance is of lower order than the squared bias. We further provide a detailed discussion on this regime in Appendix A.1. Under Assumption 3.2, we obtain the following corollary:

Corollary 3.3. *Based on Assumption 3.2 and Theorem 3.1, we further conclude that*

$$\text{Bias}(\hat{\tau}_{(1)}) = \Theta(1) \quad \text{Var}(\hat{\tau}_{(1)}) = \Theta(1/n). \quad (13)$$

This corollary shows that, for the estimator and regime considered here, the main challenge introduced by interference lies in the bias rather than the variance, especially since social networks on online platforms typically involve a large number of units. Therefore, to reduce the mean-squared error (MSE) of the estimator effectively, our primary focus should be on addressing the bias term.

3.2 Two-step Experiment

Although the derivation for $T > 2$ -step experiments does not introduce substantial additional complexity compared with the case of $T = 2$, we discuss the latter separately because the benefit of merging experimental data from two ramp-up stages with distinct treatment proportions is fundamental, compared to the single-experiment case. First, we outline the changes induced by merging two experimental datasets. For merged data, we stack the observations from different ramp-up stages and fit one OLS regression on the resulting $2n$ rows. Thus each stage contributes n observations, while the treatment proportions differ across stages. In the baseline analysis, the stage-specific noises are independent across stages and units, with common variance σ_e^2 . The treatment proportions in two experiments are c_1 and c_2 . Moreover, there are two additional changes:

$$B \xrightarrow{\text{expand}} \begin{pmatrix} B & 0 \\ 0 & B \end{pmatrix} \quad \mathbf{z} \xrightarrow{\text{expand}} \begin{pmatrix} \mathbf{z}_1 \\ \mathbf{z}_2 \end{pmatrix}. \quad (14)$$

Remark 3.4. The block-diagonal expansion in (14) rules out cross-round spillovers, as our focus is on the cross-sectional interference (see also [3]) and on estimating the static GATE. In principle, temporal interference can be incorporated by replacing $\text{diag}(B, B)$ with $\begin{pmatrix} B_1 & 0 \\ B^{(1 \rightarrow 2)} & B_2 \end{pmatrix}$, where $B^{(1 \rightarrow 2)}$ captures the linear effect of first-round treatments on second-round outcomes. Dynamic graphs can likewise be accommodated by allowing $B_1 \neq B_2$. We focus on the block-diagonal case because temporal interference changes the estimand and would blur the cross-sectional bias structure of interest. Nevertheless, our framework is flexible enough to accommodate both temporal extensions.

We then characterize the bias of the 2-step estimator $\hat{\tau}_{(2)}$ in the following theorem. In the next subsection, we will show that the variance for general $\hat{\tau}_{(T)}$ remains negligible compared to the bias under Assumption 3.2. We maintain the setting $c_1, c_2 \in (0, 1)$ and $c_1 < c_2$ as introduced in Section 2.

Theorem 3.5. *The bias of the $\hat{\tau}_{(2)}$ under general linear interference is given by:*

$$\text{Bias}(\hat{\tau}_{(2)}) = -\frac{\sum_{ij} B_{ij}}{n} \left(1 - \frac{(n-1)(c_1 - c_2)^2 + 2(c_1^2 + c_2^2) - 2(c_1 + c_2)}{(n-1)(c_1 + c_2)(2 - (c_1 + c_2))} \right). \quad (15)$$

When $n > 2(c_1 + c_2 - c_1^2 - c_2^2)/(c_1 - c_2)^2 + 1$, we have:

$$0 < \frac{(n-1)(c_1 - c_2)^2 + 2(c_1^2 + c_2^2) - 2(c_1 + c_2)}{(n-1)(c_1 + c_2)(2 - (c_1 + c_2))} < 1. \quad (16)$$

This directly implies that $\text{Bias}(\hat{\tau}_{(1)})$ and $\text{Bias}(\hat{\tau}_{(2)})$ share the same sign. A substantial reduction in the magnitude of bias from $\hat{\tau}_{(1)}$ to $\hat{\tau}_{(2)}$ is given by:

$$|\text{Bias}(\hat{\tau}_{(1)})| - |\text{Bias}(\hat{\tau}_{(2)})| = \frac{|\sum_{ij} B_{ij}|}{n} \frac{(c_1 - c_2)^2}{(c_1 + c_2)(2 - (c_1 + c_2))} + O\left(\frac{1}{n}\right). \quad (17)$$

From the perspective of the online platform, $c_1 + c_2$ can be viewed as the total budget to be allocated in the experiment. If we fix $c_1 + c_2$, increasing $|c_1 - c_2|$ corresponds to an increase in exposure variation and a decrease in bias. This bias reduction is intuitive, since estimating the GATE inherently requires estimating the two mean outcomes under treatment proportions 0 and 1, and lowering c_1 while increasing c_2 brings us closer to these target regimes, reflecting a similar intuition in [9], namely that ‘moving closer to the counterfactual situation of interest’ induces bias reduction.

Moreover, somewhat counterintuitively, merging *repeated* experiments with the same treatment proportion c does **not** substantially reduce bias, even though it increases the amount of data drawn from the same distribution. We discuss the implications of Theorem 3.5 further in Appendix A.2. Finally, since the variance analysis and conclusions for $T > 2$ closely parallel those for $T = 2$, we defer the detailed results to the next section. In particular, the variance remains of order $\Theta(1/n)$ and is therefore still negligible compared to the bias when merging T ramp-up stages.

3.3 T-step Experiment

In this section, we show that merging T ramp-up stages can still substantially reduce bias, although adding more intermediate steps may even be harmful compared to merging only the lowest and

largest treatment proportions. We then establish that the variance remains secondary relative to the bias. Here, we emphasize that our regime is one in which n is large while T remains a small fixed constant (e.g., $n = 10000$ and $T = 4$), in line with practical settings on technology platforms.

Theorem 3.6. Assume T is fixed. The bias of $\hat{\tau}_{(T)}$ under general linear interference is given by:

$$\text{Bias}(\hat{\tau}_{(T)}) = -\frac{\sum_{ij} B_{ij}}{n} \left(1 - \frac{T \sum_{t=1}^T c_t^2 - (\sum_{t=1}^T c_t)^2}{(\sum_{t=1}^T c_t)(T - \sum_{t=1}^T c_t)} \right) + O\left(\frac{1}{n}\right). \quad (18)$$

Hence, what matters is not the number of merged stages per se, but the dispersion of their treatment proportions. Repeated experiments with the same c have $V_c = 0$ and do not reduce the leading bias. Moreover, simple algebra shows that, for fixed $c_1, \dots, c_T$, merging more experimental stages does **not** necessarily lead to further bias reduction, and merging only the two extreme steps c_1 and c_T is already sufficiently effective. We defer the detailed discussion to Appendix A.3 and Appendix B.2.

Next, we analyze the variance of the regression-based estimator trained on merged data. Merging data across time introduces an additional temporal dimension in the design and may in principle create new covariance components. Nevertheless, we show that the variance remains of order $\Theta(1/n)$. We focus on two representative settings: temporally independent experiments and staggered rollout experiments [30], where treatment levels of units are non-decreasing over time.

Theorem 3.7. For the linear regression estimator trained on merged T -step data from either **temporally independent** or **staggered rollout** experiments, the order of variance is given by:

$$\text{Var}(\hat{\tau}_{(T)}) = \Theta(1/n). \quad (19)$$

4 An Intuitive Perspective on Exposure Variation

Although we study the graph-agnostic linear regression estimator for theoretical tractability, the conclusion of the previous section should extend more broadly. The key reason is that merging data across stages increases variation in macro-level treatment proportions, which **translates to variation in micro-level network exposures**. This added exposure diversity can likewise facilitate learning the interference pattern for more flexible models, such as GNN-based regressors.

Specifically, we use the case $T = 2$ and Example 2.1 for illustration. In this classical setting, the treatment exposure of unit i is defined as the fraction of its treated neighbors, $e_i = \sum_{j \in \mathcal{N}(i)} z_j / \deg_i$. We argue that the main challenge in modeling interference lies in obtaining sufficient variation in the treatment exposures $\{e_i\}_{i=1}^n$. Fortunately, because each e_i is induced by treatment randomization, the exposures naturally exhibit variation. We visualize their distribution in Figure 2.

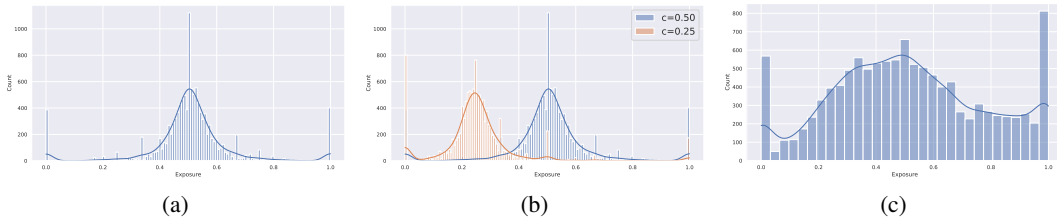


Figure 2: Fig. 2a shows the exposure distribution under treatment proportion $c = 0.5$. Fig. 2b shows the corresponding distribution with $c = 0.25, 0.5$, and Fig. 2c shows that under *cluster-level randomization* with $c = 0.5$. The exposure variances are 0.0243, 0.0377, and 0.0712, respectively.

In a single experiment under complete randomization, $\deg_i e_i$ follows a hypergeometric distribution and is well-approximated by $\text{Binom}(\deg_i, c)$ when n is large. The overall exposure distribution is therefore a mixture of these unit-level laws across i , as shown in Figure 2a. Next, Figure 2b presents two histograms for the case $c_1 = 0.25, c_2 = 0.5$. The variation in treatment exposures within the merged data increases substantially. Beyond the variation induced by randomization, merging data from experiments with different treatment proportions naturally further enriches exposure variation.

Moreover, we note that the possible harmfulness of merging intermediate steps besides the first and T -th step, as detailed in Appendix A.3, can also be explained by this principle, i.e., merging intermediate steps would in turn decrease exposure variation.

Next, Figure 2c shows the distribution of treatment exposure under cluster-level randomization [13, 26, 9], a well-established approach for causal inference under network interference in which treatment is assigned at the cluster level rather than the unit level, so that units within the same cluster (i.e., community in the network) share the same treatment level. Compared with unit-level randomization, the variation in treatment exposure becomes substantially larger, since units in the same cluster receive correlated treatments: in particular, the variance of $\{e_i\}_{i=1}^n$ increases to 0.0712. This provides another perspective on why cluster-level randomization is beneficial: by inducing covariance among treatments, it amplifies the variation in treatment exposures and thereby improves the performance of regression-based estimators. In Figure 3, we further demonstrate a significant synergistic effect between merging and cluster-level randomization, which together increase the variation in exposures. Finally, although the illustration is based on a particular exposure definition, we believe this intuition extends naturally to the most general setting, where $e_i = h_i(\mathbf{z}, A)$ with a unit-specific function h_i .

5 Simulation Study

5.1 Basic Setting

Data. We use the Facebook social network FB-Stanford3 [23], which has also been adopted in the experiments of [27, 4]. The network topology $\mathcal{G}$ contains $|\mathcal{V}| = 11586$ nodes and $|\mathcal{E}| = 568309$ edges. Based on this real network, we generate potential outcomes under the following models: linear-in-means [1], multiplicative [27], quadratic [7], multihop, and linear-in-means with dynamic graph structure. The linear-in-means and multihop models were introduced in Examples 2.1 and 2.2, respectively, and we take $M = 2$ (i.e., 2-hop) here. We defer the details of the dynamic-graph setting to Appendix B.1. We introduce the multiplicative and quadratic interference models below:

$$Y_i(\mathbf{z}) = (\beta_0 + \epsilon_i) \cdot \frac{\deg_i}{\deg} \cdot (1 + \beta_1 z_i + r \frac{\sum_{j \in \mathcal{N}_i} z_j}{\deg_i}), \quad (20)$$

$$Y_i(\mathbf{z}) = \beta_0 + \beta_1 z_i + r \left(\frac{\sum_{j \in \mathcal{N}(i)} z_j}{\deg_i} \right)^2 + \epsilon_i. \quad (21)$$

Here, $\epsilon_i \sim N(0, \sigma_e^2)$ are i.i.d. normal noises. We set the parameters as $(\beta_0, \beta_1, r, \sigma_e) = (1, 1, 1, 0.1)$.

Regression model. Throughout this section, we intentionally use a lightweight 3-layer GNN as the regression model. The details of the model architecture are reported in Appendix B.1.

Merging scheme. Next, we introduce the two merging schemes considered in our experiments. In the first, we fix the final step at $c_5 = 0.5$ and consider merging $\{c_{6-t}, \dots, c_5\}$ for $t = 1, 2, \dots, 5$. This scheme illustrates how much improvement can be achieved at $c = 0.5$ by incorporating data from previous ramp-up stages. In the second, we merge only two stages, namely $c = 1/n$ (allocating treatments to exactly one unit/cluster) and c_t for $t = 1, 2, \dots, 5$. This scheme shows how much improvement can be obtained by combining the current step with an early-stage one. This design is also motivated by Theorem 3.6, which suggests that once the lowest and highest treatment proportions are merged, incorporating additional intermediate steps may even be harmful; in other words, two-step merging is already sufficiently effective under linear interference. Throughout the experiments, we set $(c_t)_{t=1}^5 = (2\%, 5\%, 10\%, 25\%, 50\%)$.

Randomization scheme. In addition, we consider two randomization schemes: unit-level and cluster-level complete randomization, both under staggered rollout, and defer the details of cluster construction to Appendix B.1. As alternatives, independent Bernoulli randomization (vs. complete randomization) and temporally independent randomization (vs. staggered rollout) have only very limited impact on the results, as reported in Appendix B.4.

5.2 Results and Analysis

We present the main MSE results in Figure 3 and report illustrative results for multihop interference setting in Table 1. Due to space limitations, detailed numerical results and additional experiments, including alternative randomization schemes, interference patterns, estimators without regression components, and network topologies, are deferred to Appendices B.2 and B.4, respectively.

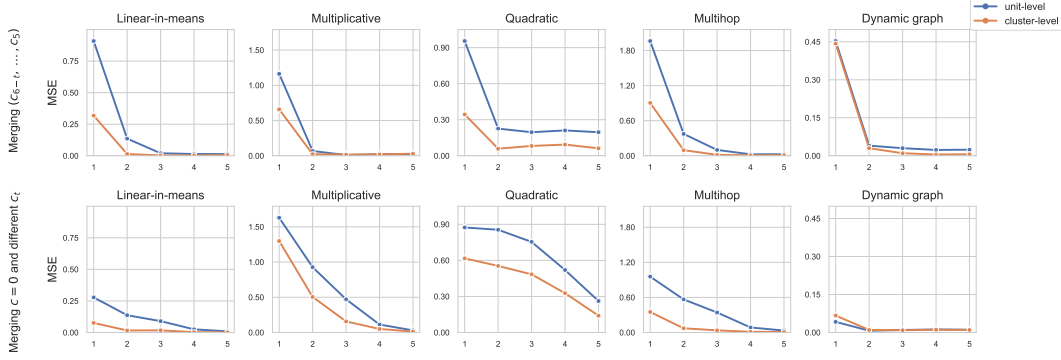


Figure 3: MSE of the GNN-based estimator under unit-level and cluster-level randomization across different interference settings. The two rows correspond to the training data formed by two merging schemes, i.e., merging $(c_{6-t}, \dots, c_5)$ and $(0, c_t)$, respectively. The x-axis indicates the timestep t .

Table 1: Performance under **2-hop** interference: (Left) merging t steps, (Right) merging $c = 0$ and c_t .

Rounds	Unit			Cluster			Proportion	Unit			Cluster		
	Bias	Std	MSE	Bias	Std	MSE		Bias	Std	MSE	Bias	Std	MSE
$t = 1$	-1.401	0.034	1.963	-0.950	0.045	0.904	$c_t = 0.02$	-0.977	0.032	0.956	-0.537	0.251	0.351
$t = 2$	-0.611	0.024	0.374	-0.303	0.064	0.096	$c_t = 0.05$	-0.751	0.043	0.566	-0.217	0.159	0.072
$t = 3$	-0.316	0.013	0.100	-0.117	0.046	0.016	$c_t = 0.10$	-0.580	0.078	0.342	-0.129	0.141	0.037
$t = 4$	-0.159	0.015	0.025	-0.061	0.064	0.008	$c_t = 0.25$	-0.293	0.020	0.086	-0.094	0.061	0.012
$t = 5$	-0.160	0.024	0.026	-0.074	0.070	0.010	$c_t = 0.50$	-0.183	0.012	0.033	-0.083	0.037	0.008

We note that the bias and standard deviation (std) are obtained from 100 Monte Carlo repetitions. We then interpret the results presented above. First, the performance of the one-step experiment with $c = 0.5$ is shown at $t = 1$ under the first merging scheme, i.e., the first column of each subfigure in the first row of Figure 3, which can be taken as a baseline for demonstrating the role of merging.

We summarize the main conclusions as follows: (i) **the benefits of two-step merging are intrinsic**; (ii) **there is a prominent synergistic effect between cluster-level randomization and our merging methodology**, as predicted in Section 4, which we attribute to their joint role in increasing exposure variation in the training data; (iii) **bias is indeed the dominant factor for the GNN-based estimator in most cases, so bias reduction substantially improves the MSE of the estimator**, e.g., see the first row of the left part of Table 1 and the tables in Appendix B.2; (iv) **merging more intermediate steps can be helpful for complex interference patterns**, e.g., non-linear interference and dynamic graph structure, see Tables 4–6; (v) the benefits of merging come from variation in treatment proportions, so merging repeated experiments is meaningless, e.g., see Appendix B.4.

Note that all these conclusions align with our derivations and discussion in Section 3 and 4, which also highlights the relevance of our theory: although exact bias–variance analysis is tractable only for linear regression, the conclusions derived there remain sufficiently general and insightful.

Overall, our comprehensive experiments suggest the following practical recommendation: when training a regression-based estimator for GATE, one should adopt cluster-level randomization and merge at least two ramp-up stages with different treatment proportions.

6 Conclusion

In this paper, we show that ramp-up is not only an operational practice but also a statistical resource for GATE estimation in A/B tests under network interference. Our closed-form analysis under general linear interference demonstrates that, in practically relevant regimes, bias is the dominant source of estimation error for regression-based estimators, and that merging experimental data with different treatment proportions can substantially reduce bias and thereby improve estimation accuracy. We attribute this improvement to the increase in exposure variation, which further predicts a strong synergy between merging and cluster-level randomization. Both insights are supported by our theory and consistently confirmed across a wide range of simulation settings.

References

- [1] Peter M Aronow and Cyrus Samii. Estimating average causal effects under general interference, with application to a social network experiment. *Annals of Applied Statistics*, 11(4):1912–1947, 2017.
- [2] Albert-László Barabási and Réka Albert. Emergence of scaling in random networks. *science*, 286(5439):509–512, 1999.
- [3] Ariel Boyarsky, Hongseok Namkoong, and Jean Pouget-Abadie. Modeling interference using experiment roll-out. In *Proceedings of the 24th ACM Conference on Economics and Computation*, EC ’23, 2023.
- [4] Qianyi Chen, Bo Li, Lu Deng, and Yong Wang. Optimized covariance design for ab test on social network under interference. In *Advances in Neural Information Processing Systems*, 2023.
- [5] Qianyi Chen, Anpeng Wu, Bo Li, Lu Deng, and Yong Wang. Journey to the centre of cluster: Harnessing interior nodes for a/b testing under network interference. In *The Fourteenth International Conference on Learning Representations*, 2026.
- [6] Victor Chernozhukov, Denis Chetverikov, Mert Demirer, Esther Duflo, Christian Hansen, Whitney Newey, and James Robins. Double/debiased machine learning for treatment and structural parameters. *The Econometrics Journal*, 21(1):C1–C68, 01 2018.
- [7] Mayleen Cortez, Matthew Eichhorn, and Christina Yu. Staggered rollout designs enable causal inference under interference without network knowledge. In *Advances in Neural Information Processing Systems*, 2022.
- [8] Michaël Defferrard, Xavier Bresson, and Pierre Vandergheynst. Convolutional neural networks on graphs with fast localized spectral filtering. *Advances in neural information processing systems*, 29, 2016.
- [9] Dean Eckles, Brian Karrer, and Johan Ugander. Design and analysis of experiments in networks: Reducing bias from interference. *Journal of Causal Inference*, 5(1):20150021, 2016.
- [10] Laura Forastiere, Fabrizia Mealli, Albert Wu, and Edoardo M Airoidi. Estimating causal effects under network interference with bayesian generalized propensity scores. *Journal of Machine Learning Research*, 23(289):1–61, 2022.
- [11] Mengsi Gao and Peng Ding. Causal inference in network experiments: regression-based analysis and design-based properties. *Journal of Econometrics*, 252:106119, 2025.
- [12] Qiang Huang and Jin Tian. Modeling interference for treatment effect estimation in network dynamic environment. In *The Fourteenth International Conference on Learning Representations*, 2026.
- [13] Michael G Hudgens and M Elizabeth Halloran. Toward causal inference with interference. *Journal of the American Statistical Association*, 103(482):832–842, 2008.
- [14] Su Jia, Nathan Kallus, and Christina Lee Yu. Clustered switchback experiments: Near-optimal rates under spatiotemporal interference. *arXiv preprint arXiv:2312.15574*, 2025.
- [15] Vardis Kandiros, Charilaos Pipis, Constantinos Daskalakis, and Christopher Harshaw. The conflict graph design: Estimating causal effects under arbitrary neighborhood interference. *arXiv preprint arXiv:2411.10908*, 2026.
- [16] Diederik P Kingma and Jimmy Ba. Adam: A method for stochastic optimization. *arXiv preprint arXiv:1412.6980*, 2017.
- [17] Ron Kohavi, Diane Tang, and Ya Xu. *Trustworthy online controlled experiments: A practical guide to a/b testing*. Cambridge University Press, 2020.
- [18] Jure Leskovec, Jon Kleinberg, and Christos Faloutsos. Graphs over time: densification laws, shrinking diameters and possible explanations. In *Proceedings of the eleventh ACM SIGKDD international conference on Knowledge discovery in data mining*, pages 177–187, 2005.
- [19] Michael P Leung. Causal inference under approximate neighborhood interference. *Econometrica*, 90(1):267–293, 2022.
- [20] Yang Liu, Yifan Zhou, Ping Li, and Feifang Hu. Cluster-adaptive network a/b testing: From randomization to estimation. *Journal of Machine Learning Research*, 25(170):1–48, 2024.

- 391 [21] Ben M Parker, Steven G Gilmour, and John Schormans. Optimal design of experiments on
392 connected units with application to social networks. *Journal of the Royal Statistical Society.*
393 *Series C (Applied Statistics)*, pages 455–480, 2017.
- 394 [22] Paul R Rosenbaum. Interference between units in randomized experiments. *Journal of the*
395 *American Statistical Association*, 102(477):191–200, 2007.
- 396 [23] Ryan Rossi and Nesreen Ahmed. The network data repository with interactive graph analytics
397 and visualization. In *Proceedings of the AAAI conference on artificial intelligence*, 2015.
- 398 [24] Donald B Rubin. Estimating causal effects of treatments in randomized and nonrandomized
399 studies. *Journal of educational Psychology*, 66(5):688, 1974.
- 400 [25] Michael E Sobel. What do randomized studies of housing mobility demonstrate? causal infer-
401 ence in the face of interference. *Journal of the American Statistical Association*, 101(476):1398–
402 1407, 2006.
- 403 [26] Johan Ugander, Brian Karrer, Lars Backstrom, and Jon Kleinberg. Graph cluster randomization:
404 Network exposure to multiple universes. In *Proceedings of the 19th ACM SIGKDD international*
405 *conference on Knowledge discovery and data mining*, 2013.
- 406 [27] Johan Ugander and Hao Yin. Randomized graph cluster randomization. *Journal of Causal*
407 *Inference*, 11(1):20220014, 2023.
- 408 [28] Minjie Wang, Da Zheng, Zihao Ye, Quan Gan, Mufei Li, Xiang Song, Jinjing Zhou, Chao Ma,
409 Lingfan Yu, Yu Gai, et al. Deep graph library: A graph-centric, highly-performant package for
410 graph neural networks. *arXiv preprint arXiv:1909.01315*, 2020.
- 411 [29] Anpeng Wu, Haiyi Qiu, Zhengming Chen, Zijian Li, Ruoxuan Xiong, Fei Wu, and Kun Zhang.
412 Causal graph transformer for treatment effect estimation under unknown interference. In *The*
413 *Thirteenth International Conference on Learning Representations*, 2025.
- 414 [30] Ruoxuan Xiong, Susan Athey, Mohsen Bayati, and Guido Imbens. Optimal experimental design
415 for staggered rollouts. *Management Science*, 70(8):5317–5336, 2023.
- 416 [31] Ya Xu, Weitao Duan, and Shaochen Huang. Sqr: Balancing speed, quality and risk in online
417 experiments. In *Proceedings of the 24th ACM SIGKDD International Conference on Knowledge*
418 *Discovery & Data Mining*, pages 895–904, 2018.

A Additional Discussion on Theoretical Results

A.1 Discussion on Assumption 3.2

In this subsection, we revisit Assumption 3.2 and discuss its implications and practicality.

Scenarios where interference is excessively strong are relatively rare. Moreover, to assess the relative order of variance compared to bias in such cases, we will inevitably encounter the order of $\sum_{(i,j,k,l) \in \mathcal{I}} B_{ij} B_{kl} / \sum_{i,j} B_{ij}$ for various index sets $\mathcal{I}$ as outlined in Theorem 3.1. This complexity makes it challenging to derive succinct conclusions without specifying the interference matrix B .

For a specific unit i , the interference on its outcome is represented by $\sum_j B_{ij} z_j$, and the treatment effect is β_1 . The first part of Assumption 3.2 restricts the interference effect received by unit i to be at most of the constant order, ensuring that interference does not overshadow the outcome. Additionally, this assumption allows for some units to be minimally affected by interference. Finally, through basic algebra, one can verify that $\sum_j |B_{ij}|$ is equal to 1 in Example 2.1 and bounded above by $\sum_{m=1}^M r_m$ in Example 2.2.

The second part of Assumption 3.2 restricts the order of total interference effect to be regular. While we assume that $\sum_{i,j} B_{ij} = \Theta(n)$, the bias reduction results discussed in the subsequent subsections also apply when $\sum_{i,j} B_{ij} = o(n)$, which indicates very weak interference. We do not incorporate this case here because the bias introduced by interference diminishes as n increases, given that $\sum_{i,j} B_{ij}/n = o(1)$. As illustrated in the expression of bias, it is unnecessary to use specific techniques designed to address interference, in this case. Therefore, we focus on the meaningful and challenging scenario, where $\sum_{i,j} B_{ij} = \Theta(n)$. For instance, in Examples 2.1 and 2.2, the order of $\sum_{ij} B_{ij}$ are both $\Theta(n)$.

The final part of Assumption 3.2 restricts the order of the sum of the total impact that unit i imposes on other units. Intuitively, it restricts the number and influence of highly influential opinion leaders within the network. In Example 2.1, this assumption does not always hold and thus imposes substantial constraints on the adjacency matrix A , specifically, $\mathbf{1}^\top D^{-1} A^2 D^{-1} \mathbf{1} = O(n)$.

For instance, in a ring graph, we have $\sum_j (\sum_i B_{ij})^2 = \Theta(n)$, while in a star graph—where one opinion leader connects to all other units—we have $\sum_j (\sum_i B_{ij})^2 = \Theta(n^2)$. In the context of Example 2.1, a sufficient condition for satisfying this assumption is the restricted growth rate that is commonly employed in related works [26, 27, 14]. However, a restricted growth rate is not a necessary condition; e.g. $\mathbf{1}^\top D^{-1} A^2 D^{-1} \mathbf{1} = n$ in the complete graph; despite being the densest graph, it lacks any prominent opinion leaders. This assumption is particularly reasonable for typical social networks, where, even with over a billion nodes, it is unlikely for any single unit to have thousands of friends while maintaining significant influence over each one of them.

Lastly, the final part of Assumption 3.2 is not required for deriving the Corollary 3.3 but is instead intended to support the conclusion in the case of merging data, specifically that the variance of the linear regression estimator remains at the order of $\Theta(1/n)$.

A.2 Discussion on Theorem 3.5

Lowering c_1 and increasing c_2 brings us closer to these target mean outcomes. Another way to roughly characterize the budget is to ensure $\max\{c_1, c_2\} \leq c$ since a higher treatment proportion introduces significantly more risk to revenue and/or user experience. In broad terms, the most effective way to reduce bias is setting $c_1 = 0, c_2 = c$. However, even if we set $c_1 = 0, c_2 = c$, the bias becomes:

$$-\frac{\sum_{ij} B_{ij}}{n} \left(1 - \frac{c}{2-c}\right) + O\left(\frac{1}{n}\right), \quad (22)$$

which is still numerically unsatisfactory, particularly since $c \leq 0.5$ is common in practice. Moreover, as demonstrated in the next subsection, merging multistep experimental data does not lead to further bias reduction compared to the two-step case. Therefore, we choose to explore more powerful regression models, specifically GNNs, in our simulation study.

464 A.3 Discussion on Theorem 3.6

465 In machine learning practice, it is common to see that merging more related data enhances the
 466 performance of the predictive models. However, through basic algebra, one can verify that merging
 467 more experimental data does **not** guarantee further bias reduction. This raises a natural question:
 468 Given the completed experiments with treatment proportions $c_1 < c_2 < \dots < c_T$, which new
 469 experiment with proportion x will further reduce bias? We address this question through the following
 470 corollary.

471 **Corollary A.1.** *To further reduce the bias, x should satisfy:*

$$(T^2C_1 - (T+1)C_1^2 + TC_2)x^2 - ((T-1)C_1^2 + 2TC_1C_2 + T(T+1)C_2)x + C_1^3 - C_1^2C_2 > 0, \quad (23)$$

472 here $C_1 = \sum_{t=1}^T c_t$, $C_2 = \sum_{t=1}^T c_t^2$

473 For this quadratic function with respect to x , we notice the following two facts about the coefficient
 474 of the quadratic term and the constant term:

$$\begin{aligned} T^2C_1 - (T+1)C_1^2 + TC_2 &= (TC_1(T - C_1)) + (TC_2 - C_1^2) > 0 \\ C_1^3 - C_1^2C_2 &= C_1^2(C_1 - C_2) > 0. \end{aligned} \quad (24)$$

475 We then draw several conclusions about the quadratic function in Corollary A.1. First, when $c_1 > 0$,
 476 merging data with $x = 0$ —representing data under global control—can always reduce bias. In
 477 practice, this type of data is almost always accessible on online platforms. Second, intuitively, a
 478 beneficial proportion x should either be lower than c_1 or higher than c_T , rather than lying within the
 479 middle section of $[c_1, c_T]$. This intuition is only for illustration and not rigorous, since x around c_1 or
 480 c_T may also bring slight bias reduction, which depends on the specific values of $(c_t)_{t=1}^T$. Considering
 481 the first point discussed above, an effective way to further reduce bias is to conduct a larger-scale
 482 experiment, i.e., with $x > c_T$.

483 Additionally, we can apply Corollary A.1 in reverse: given the treatment proportions $\{c_i\}_{i=1}^T/\{c_t\}$,
 484 removing data with the treatment proportion c_t from the merged dataset may further reduce bias.
 485 Repeating this procedure suggests that the optimal solution may involve merging only the datasets
 486 with proportions c_1 and c_T . This finding highlights the unique importance of the $T = 2$ case, which
 487 we will further validate in the simulation study. Nonetheless, with nonlinear interference terms,
 488 merging multistep experimental data can offer advantages, as will be discussed in Section 5.

489 While it is theoretically possible to solve an optimization problem to select the treatment proportions
 490 $(c_t)_{t=1}^T$ for maximizing bias reduction, we do not recommend this approach. First, the optimization
 491 problem is dependent on the specific potential outcome model and the regression model used.
 492 Although we provide a detailed analysis of the simple regression estimator, it is not typically used in
 493 practice. Second, when selecting ramp-up steps in real-world experiments, the experimenter must
 494 consider factors such as risk and budget in advance, while such an optimization problem does not
 495 capture these practical considerations.

496 Finally, we discuss our proposed intuition of increasing exposure variation, as introduced in Sec-
 497 tion 4. For example, the variance of treatment exposures of merged data with treatment proportions
 498 (10%, 50%) is 0.0587. However, when we insert another ramp-up stage with $c = 0.25$, the variance of
 499 treatment exposures decreases to 0.0459. This corresponds to an increase in bias when experimental
 500 data with a treatment proportion x lying in the middle range of $[c_1, c_T]$ is further merged. Similarly,
 501 the maximum variance is achieved when exactly 50% treatment exposures are at 0 and 1, respectively.
 502 This is the optimal scenario for GATE estimation, albeit impractical.

503 A.4 Discussion on Temporal Interference

504 The focus of this paper is network interference, namely the cross-sectional component. Although
 505 temporal interference may arise across multiple experiments, it is not our primary concern, and
 506 we abstract it away in our analysis. In principle, our analytical framework can be extended to
 507 accommodate both temporal interference and dynamic graph structures; however, doing so would
 508 alter the target estimand—the static GATE—and complicate the main narrative of this paper. In
 509 short, we study the regime in which cross-sectional interference is more substantial than temporal
 510 interference and therefore deserves primary attention.

B Detailed Experiments Results

B.1 Details of Basic Setting

Data. The main network topology is a Facebook social network used in many related works, available at <https://networkrepository.com/socfb-Stanford3.php>.

GNN. We choose the Chebyshev Spectral Graph Convolution (ChebConv) [8] as our regression model, implemented through the Deep Graph Library [28]. Our GNN model consists of three ChebConv layers, with input dimension, output dimension, and filter size (2, 16, 2), (16, 16, 1), (16, 1, 1), respectively.

We optimize the GNN using the Adam optimizer [16] and MSE as the loss function. In the training process, we ensure that the GNN model is adequately optimized across most cases. This is indicated by the fact that the MSE loss converges to the level of σ_e^2 , i.e. the irreducible variance introduced by exogenous noise.

Specifically, we only select the treatment of unit z_i and its degree $\deg_i$ as initial features for each unit i . Other miscellaneous parameters are given below:

1. Resolution of clustering: 10
2. Number of clusters: 192
3. Activation function: ReLU
4. Number of training epochs: 400
5. Optimizer: Adam with learning rate 0.004 and momentum coefficient (betas) 0.9, 0.999.

Potential outcome models. Beyond the results of the standard linear-in-means model (Example 2.1), we examine three non-linear interference patterns to demonstrate the generality of our theoretical insights, including **multiplicative**, **quadratic**, and **square-root interference**.

$$Y_i(\mathbf{z}) = (\beta_0 + \epsilon_i) \cdot \frac{\deg_i}{\deg} \cdot (1 + \beta_1 z_i + r \frac{\sum_{j \in \mathcal{N}_i} z_j}{\deg_i}) \quad (25)$$

$$Y_i(\mathbf{z}) = \beta_0 + \beta_1 z_i + r \left(\frac{\sum_{j \in \mathcal{N}(i)} z_j}{\deg_i} \right)^2 + \epsilon_i \quad (26)$$

$$Y_i(\mathbf{z}) = \beta_0 + \beta_1 z_i + r \sqrt{\left(\frac{\sum_{j \in \mathcal{N}(i)} z_j}{\deg_i} \right)} + \epsilon_i, \quad (27)$$

Here, $\epsilon_i \sim N(0, \sigma_e^2)$ are i.i.d. normal noises. We set the parameters as $(\beta_0, \beta_1, r, \sigma_e) = (1, 1, 1, 0.1)$. We defer the introduction of our dynamic graph structure to Appendix B.3.

B.2 Detailed Results

We present the main numerical results below. Table 2 shows that the theoretical conclusions derived in Section 3 and Appendix A continue to hold for the GNN regressor. In particular, merging the first and last ramp-up stages (c_1, c_T) is already highly effective, and can even outperform merging intermediate stages. Under more complex interference patterns, however, incorporating additional intermediate stages can be substantially more beneficial, much like two points suffice to determine a line, whereas a cubic curve requires four.

B.3 Dynamic Graph Structure

We introduce our setting of a dynamic graph structure, which is novel in the literature but common in practice. For instance, if each experiment is conducted over a period of one week, the edges of the graph can change within that week, even in social networks. New friendships may form, and existing ones may be dissolved. The evolution of network structure is well-documented in network science, with models such as the Barabási-Albert Model [2] and the Forest Fire Model [18]. Although this

Table 2: Performance under **linear-in-means** model (1-hop linear interference): (Left) merging t steps, (Right) merging $c = 0$ and c_t .

Rounds	Unit			Cluster			Proportion	Unit			Cluster		
	Bias	Std	MSE	Bias	Std	MSE		Bias	Std	MSE	Bias	Std	MSE
$t = 1$	-0.951	0.071	0.909	-0.561	0.059	0.318	$c_t = 0.02$	-0.526	0.034	0.278	-0.216	0.173	0.076
$t = 2$	-0.369	0.021	0.136	-0.112	0.037	0.014	$c_t = 0.05$	-0.369	0.025	0.137	-0.032	0.121	0.016
$t = 3$	-0.142	0.014	0.020	-0.034	0.044	0.003	$c_t = 0.10$	-0.299	0.028	0.090	-0.010	0.130	0.017
$t = 4$	-0.116	0.023	0.014	-0.030	0.048	0.003	$c_t = 0.25$	-0.155	0.020	0.025	-0.037	0.045	0.003
$t = 5$	-0.110	0.020	0.013	-0.042	0.054	0.005	$c_t = 0.50$	-0.092	0.015	0.009	-0.015	0.033	0.001

Table 3: Performance with **multiplicative** model: (Left) merging t steps, (Right) merging $c = 0$ and c_t .

Rounds	Unit			Cluster			Proportion	Unit			Cluster		
	Bias	Std	MSE	Bias	Std	MSE		Bias	Std	MSE	Bias	Std	MSE
$t = 1$	-1.078	0.039	1.163	-0.811	0.046	0.659	$c_t = 0.02$	-1.275	0.055	1.629	-1.103	0.288	1.299
$t = 2$	-0.257	0.037	0.068	-0.122	0.089	0.023	$c_t = 0.05$	-0.959	0.082	0.927	-0.666	0.249	0.505
$t = 3$	-0.061	0.077	0.010	-0.058	0.115	0.017	$c_t = 0.10$	-0.681	0.082	0.471	-0.340	0.205	0.157
$t = 4$	-0.110	0.106	0.023	-0.003	0.137	0.019	$c_t = 0.25$	-0.321	0.098	0.113	-0.193	0.111	0.050
$t = 5$	-0.120	0.104	0.025	-0.118	0.125	0.030	$c_t = 0.50$	-0.164	0.058	0.030	-0.101	0.063	0.014

Table 4: Performance with **quadratic** interference term: (Left) merging t steps, (Right) merging $c = 0$ and c_t .

Rounds	Unit			Cluster			Proportion	Unit			Cluster		
	Bias	Std	MSE	Bias	Std	MSE		Bias	Std	MSE	Bias	Std	MSE
$t = 1$	-0.976	0.044	0.955	-0.583	0.063	0.344	$c_t = 0.02$	-0.932	0.067	0.874	-0.776	0.120	0.617
$t = 2$	-0.475	0.020	0.226	-0.240	0.040	0.059	$c_t = 0.05$	-0.924	0.032	0.855	-0.739	0.084	0.554
$t = 3$	-0.443	0.010	0.196	-0.284	0.038	0.082	$c_t = 0.10$	-0.868	0.021	0.754	-0.692	0.074	0.484
$t = 4$	-0.460	0.015	0.211	-0.301	0.044	0.093	$c_t = 0.25$	-0.721	0.023	0.520	-0.569	0.058	0.327
$t = 5$	-0.442	0.032	0.196	-0.241	0.064	0.062	$c_t = 0.50$	-0.512	0.010	0.262	-0.373	0.035	0.140

Table 5: Performance with **square-root** interference term: (Left) merging t steps, (Right) merging $c = 0$ and c_t .

Rounds	Unit			Cluster			Proportion	Unit			Cluster		
	Bias	Std	MSE	Bias	Std	MSE		Bias	Std	MSE	Bias	Std	MSE
$t = 1$	-0.905	0.042	0.821	-0.623	0.051	0.391	$c_t = 0.02$	-0.324	0.027	0.106	0.218	0.304	0.140
$t = 2$	-0.387	0.014	0.150	-0.180	0.046	0.035	$c_t = 0.05$	-0.090	0.109	0.020	0.696	0.241	0.543
$t = 3$	-0.185	0.012	0.034	-0.063	0.052	0.007	$c_t = 0.10$	0.519	0.050	0.272	0.895	0.145	0.822
$t = 4$	-0.086	0.010	0.008	0.021	0.057	0.004	$c_t = 0.25$	0.528	0.030	0.280	0.579	0.091	0.344
$t = 5$	-0.006	0.012	0.000	0.030	0.081	0.007	$c_t = 0.50$	0.218	0.008	0.048	0.250	0.057	0.066

Table 6: Performance with **dynamic graph structure**: (Left) merging t steps, (Right) merging $c = 0$ and c_t .

Rounds	Unit			Cluster			Proportion	Unit			Cluster		
	Bias	Std	MSE	Bias	Std	MSE		Bias	Std	MSE	Bias	Std	MSE
$t = 1$	-0.666	0.095	0.453	-0.665	0.025	0.443	$c_t = 0.02$	-0.196	0.061	0.042	-0.220	0.135	0.067
$t = 2$	-0.150	0.132	0.040	-0.161	0.061	0.030	$c_t = 0.05$	-0.069	0.048	0.007	-0.086	0.055	0.010
$t = 3$	-0.106	0.137	0.030	-0.094	0.028	0.010	$c_t = 0.10$	-0.092	0.037	0.010	-0.084	0.041	0.009
$t = 4$	-0.060	0.141	0.023	-0.058	0.044	0.005	$c_t = 0.25$	-0.107	0.021	0.012	-0.100	0.025	0.011
$t = 5$	-0.045	0.149	0.024	-0.034	0.070	0.006	$c_t = 0.50$	-0.099	0.031	0.011	-0.098	0.028	0.010

548 aspect is still underexplored in the context of network experiments, it is valuable to consider this
549 practical factor in sequential multiple experiments, such as in the ramp-up process.

550 *Remark B.1.* Here, for simplicity, we consider the exogenous network evolution that is independent
551 of treatment assignment. This is because the treatments typically considered by online platforms, such
552 as new product features, generally do not have a significant impact on social relationships. However,
553 there is existing literature in economics that examines cases where treatment does endogenously
554 influence network formation, usually when the treatment itself is more consequential.

555 For simplicity, we keep the node set unchanged throughout the ramp-up process and focus on
556 modeling the addition of new edges over time. Inspired by classic preferential attachment, we
557 generate edges for nodes with degrees in the top 50%. For each such node i , we connect it to node j
558 with the following probability:

$$p_{ij} = \frac{\deg_j}{\sum_k \deg_k}, \quad (28)$$

In each evolution, we repeat this sampling procedure 5 times with replacement for each node with degrees in the top 50%. If a unit i is sampled when generating edges for itself, we discard this sampling to avoid self-loops. Additionally, if the sampled unit j is already connected to unit i at the current step, we also discard this sampling to avoid creating multiple edges.

Given the ramp-up steps $(c_1, c_2, \dots, c_5) = (2\%, 5\%, 10\%, 25\%, 50\%)$, we take the initial network topology $\mathcal{G}$ as baseline and repeat such evolution 5 times to generate evolved graphs with the same node set. The number of edges:

$$|\mathcal{E}_1|, \dots, |\mathcal{E}_5| = 596414, 624465, 652391, 680313, 708274, \quad (29)$$

where $\mathcal{E}_t$ is the edge set at the t -th step.

Moreover, we go back to use the classic linear-in-means model and examine the influence of dynamic graph structure. The main parameter is set as $(\beta_0, \beta_1, r, \sigma_e) = (1, 1, 1, 0.1)$, as in the previous settings. Even if the graph structure evolves, the GATE remains $\beta_1 + r = 2$. We present the result in Table 6.

The impact of evolutionary changes is quite complex, with classic models typically only characterizing the limiting behavior of preferential attachment. Nonetheless, our methodology consistently enhances the GNN model’s performance. During the evolution process, we preserve the clustering of the original topology $\mathcal{G}$ to implement the staggered rollout design¹, although this clustering may become less effective as edges are continuously added. As a result, our suggested practice of merging data with global control and the highest treatment proportion faces challenges due to the increasing discrepancy between $\mathcal{G}_t$ and $\mathcal{G}$ as step t progresses. This provides an intuitive explanation for why the monotonic improvement observed with increasing c_t , as shown in our basic experiments, no longer persists in Table 6. This reveals a potential limitation of this merging strategy in practice.

Remark B.2. The rationale for generating edges solely for high-degree nodes stems from the distribution of treatment exposure, as detailed in the main paper. Low-degree nodes typically have exposure values concentrated at the boundaries of the interval $(0, 1)$, which significantly contributes to the variation in exposures. To preserve this distinctive structure, we intentionally avoid generating new edges for all nodes. However, we do allow low-degree nodes to receive new edges initiated by high-degree nodes.

B.4 Supplementary Experiments

Randomization issue. We note that the independent Bernoulli randomization is used more widely in industrial practice, due to its simplicity. However, the correlation among treatments introduced in complete randomization is actually also very weak, so they perform very similarly on most cases. Moreover, we also supplement a comparison of the temporal aspect of randomization, i.e., temporally independent experiments vs. staggered rollout experiments, which forms a 2×2 setting grid. The results are presented in Table 7 and Table 8.

Table 7: Performance under **independent Bernoulli randomization**: (Left) **temporal independence**, (Right) **staggered rollout**.

Rounds	Unit			Cluster			Rounds	Unit			Cluster		
	Bias	Std	MSE	Bias	Std	MSE		Bias	Std	MSE	Bias	Std	MSE
$t = 1$	-0.945	0.071	0.899	-0.562	0.058	0.320	$t = 1$	-0.944	0.072	0.896	-0.560	0.058	0.317
$t = 2$	-0.365	0.023	0.133	-0.118	0.046	0.016	$t = 2$	-0.365	0.022	0.134	-0.117	0.047	0.016
$t = 3$	-0.142	0.015	0.020	-0.038	0.041	0.003	$t = 3$	-0.142	0.015	0.020	-0.036	0.044	0.003
$t = 4$	-0.115	0.024	0.014	-0.032	0.050	0.004	$t = 4$	-0.116	0.023	0.014	-0.033	0.051	0.004
$t = 5$	-0.110	0.019	0.012	-0.042	0.056	0.005	$t = 5$	-0.110	0.019	0.012	-0.042	0.056	0.005

Merging repeated experiments. Given the significant improvement brought by merging experimental data, a natural question arises: Is the observed improvement solely due to the increased volume of training data? In Theorem 3.5 and 3.6, we conclude that the potential bias reduction of merging experiments with the same treatment proportion should be very marginal, i.e., at most of order $O(1/n)$, and we examine this for the linear regression estimator here.

¹The time complexity of the Louvain algorithm supports weekly updates for clustering on large online platforms. However, when clustering changes, monotonic treatment assignment may no longer be guaranteed under cluster-level randomization.

Table 8: Performance under **complete randomization**: (Left) **temporal independence**, (Right) **staggered rollout**.

Rounds	Unit			Cluster			Rounds	Unit			Cluster		
	Bias	Std	MSE	Bias	Std	MSE		Bias	Std	MSE	Bias	Std	MSE
$t = 1$	-0.953	0.071	0.914	-0.560	0.059	0.317	$t = 1$	-0.951	0.071	0.909	-0.561	0.059	0.318
$t = 2$	-0.369	0.021	0.136	-0.114	0.037	0.014	$t = 2$	-0.369	0.021	0.136	-0.112	0.037	0.014
$t = 3$	-0.141	0.015	0.020	-0.033	0.045	0.003	$t = 3$	-0.142	0.014	0.020	-0.034	0.044	0.003
$t = 4$	-0.115	0.024	0.014	-0.031	0.047	0.003	$t = 4$	-0.116	0.023	0.014	-0.030	0.048	0.003
$t = 5$	-0.110	0.020	0.013	-0.042	0.054	0.005	$t = 5$	-0.110	0.020	0.013	-0.042	0.054	0.005

Table 9 presents the results under 2-hop interference and merged t repeated experiments, with the treatment proportion for each experiment fixed at 50%. We observe that the benefits of merging data from these repeated experiments are minimal. Ultimately, we conclude that substantial bias reduction is achieved primarily by merging experimental data with **different** treatment proportions. This aligns with our theoretical findings for the linear regression estimator detailed in Theorems 3.5 and 3.6.

Table 9: Performance under 2-hop interference with **merged repeated experiments over t steps**.

Rounds	Unit			Cluster		
	Bias	Std	MSE	Bias	Std	MSE
$t = 1$	-1.402	0.034	1.968	-0.954	0.049	0.912
$t = 2$	-1.460	0.015	2.132	-0.981	0.028	0.963
$t = 3$	-1.480	0.012	2.189	-0.992	0.025	0.984
$t = 4$	-1.486	0.012	2.208	-0.993	0.025	0.987
$t = 5$	-1.488	0.012	2.214	-0.994	0.026	0.989

Linear regression estimator We then supplement the statistical performance of the linear regression estimator under the basic setting, as shown in Figure 1. As in the main paper, we fix the randomization scheme as complete randomization with staggered rollout. We consider treatment proportions $(c_1, c_2, \dots, c_5) = (2\%, 5\%, 10\%, 25\%, 50\%)$ in the ramp-up process and train on merged data from $t = 1, 2, 3, 4, 5$ steps, corresponding to treatment proportions $(c_{6-t}, \dots, c_5)$. The detailed results are reported in Table 10. By basic algebra, these numerical results validate Theorem 3.5.

Table 10: Performance with **linear regression estimator** when merging t steps.

Rounds	Unit			Cluster		
	Bias	Std	MSE	Bias	Std	MSE
$t = 1$	-1.000	0.004	1.000	-0.595	0.010	0.354
$t = 2$	-0.933	0.003	0.871	-0.554	0.019	0.307
$t = 3$	-0.866	0.003	0.750	-0.518	0.027	0.269
$t = 4$	-0.824	0.003	0.680	-0.493	0.028	0.244
$t = 5$	-0.792	0.003	0.628	-0.471	0.031	0.223

Lagrange interpolation estimator We next supplement the performance of the Lagrange interpolation estimator proposed in [7]. Unlike our regression-based approach, this is a system-level estimator that is agnostic to the graph microstructure: it approximates the one-dimensional mean outcome curve $\bar{Y}(p)$ as a function of the treatment proportion p under unit-level independent Bernoulli randomization. Its polynomial basis effectively assumes a low-order interaction structure of the form

$$\sum_{\ell=1}^{\beta} w_{\ell} \left(\frac{\sum_{j \in \mathcal{N}_i} \tilde{c}_{ij} z_j}{\sum_{j \in \mathcal{N}_i} \tilde{c}_{ij}} \right)^{\ell}, \quad (30)$$

where $\tilde{c}_{ij}$ characterizes the influence of unit j on unit i , and w_{ℓ} is the coefficient of the ℓ -th order term.

We note that Lagrange interpolation estimator is a **graph-agnostic** estimator that summarizes the experimental data at each timestep into a single value—mean outcomes. Thus, it heavily relies on the exact knowledge of β and incurs misspecification risk for both potential outcome model and **number of timesteps** (note that the order of polynomial in Lagrange interpolation is often exactly $T - 1$ due to random error). We report its performance on two settings: multiplicative interference in Table 11 and square-root interference in Table 12.

In the multiplicative setting, we observe that Lagrange interpolation performs very well for the case of $t = 2$, while it rapidly deteriorates with the increase of t , which is counterintuitive but can be explained by our summary: **it heavily relies on the exact knowledge on β** , which is usually inaccessible in practice. Moreover, although the theoretical derivation of both our paper and [7] are based on unit-level randomization, the functional form of Lagrange interpolation estimator there substantially relies on the randomization, while our idea is generalizable (through the lens of increasing exposure variation), i.e., improving the data for learning representation. Therefore, we observe a general synergistic effect with cluster-level randomization and our merging methodology, while Lagrange interpolation estimator performs even much worse on experimental data generated with cluster-level randomization.

In the square-root setting (the performance of GNN-based estimator with our methodology is provided in Table 5), which corresponds a scenario of diminishing marginal utility, our methodology benefits a lot from merging more steps (left part of Table 5), while

We summarize that our GNN-based approach learns a flexible node-level representation from all units directly, and can naturally accommodate richer microstructure, nodal covariates, and more complex interaction patterns.

Table 11: Performance of **Lagrange interpolation estimator** with t steps and **multiplicative** interference.

Rounds	Unit			Cluster		
	Bias	Std	MSE	Bias	Std	MSE
$t = 1$	N/A	N/A	N/A	N/A	N/A	N/A
$t = 2$	-0.002	0.036	0.001	0.010	0.237	0.056
$t = 3$	-0.005	0.074	0.005	0.045	0.479	0.231
$t = 4$	0.013	0.561	0.315	0.740	3.501	12.806
$t = 5$	0.045	8.437	71.192	4.802	49.852	2508.301

Table 12: Performance of **Lagrange interpolation estimator** with t steps and **square-root** interference.

Rounds	Unit			Cluster		
	Bias	Std	MSE	Bias	Std	MSE
$t = 1$	N/A	N/A	N/A	N/A	N/A	N/A
$t = 2$	-0.131	0.015	0.017	-0.079	0.331	0.116
$t = 3$	-0.387	0.032	0.150	-0.282	0.685	0.549
$t = 4$	1.276	0.366	1.761	1.787	5.708	35.769
$t = 5$	-18.021	7.106	375.254	-8.481	88.365	7880.220

Network-aware estimators without regression component. We next provide the performance of two network-aware estimators without regression component for comparison, including MII [5] and Hájek estimator [27], in Table 13. We empirically find that naively averaging the estimates acquired on different timesteps t brings no improvement, so we only report the performance with treatment proportion $t = 50\%$. This table is provided only for comparison.

Table 13: Performance of MII and Hájek estimators under 2-hop interference.

Estimator	Unit			Cluster		
	Bias	Std	MSE	Bias	Std	MSE
MI	N/A	N/A	N/A	-0.276	0.010	0.077
Hájek	-0.471	0.043	0.223	-0.263	0.039	0.071

Another network topology. For completeness, we supplement the performance of GNN-based estimator on MIT8² [23]. We examine the case of 2-hop interference, since it is concerned with the concrete topology. We report the results in Table 14, and the results demonstrate that our methodology remains effective.

²The network topology is available at <https://networkrepository.com/socfb-MIT8.php>.

Table 14: Performance under **2-hop** interference on the **MIT8**: (Left) merging t steps, (Right) merging $c = 0$ and c_t .

Rounds	Unit			Cluster			Proportion	Unit			Cluster		
	Bias	Std	MSE	Bias	Std	MSE		Bias	Std	MSE	Bias	Std	MSE
$t = 1$	-1.439	0.023	2.072	-1.022	0.025	1.044	$c_t = 0.02$	-0.880	0.052	0.776	-0.467	0.256	0.284
$t = 2$	-0.729	0.018	0.531	-0.411	0.052	0.172	$c_t = 0.05$	-0.633	0.052	0.403	-0.246	0.119	0.075
$t = 3$	-0.351	0.018	0.123	-0.170	0.033	0.030	$c_t = 0.10$	-0.545	0.032	0.298	-0.192	0.055	0.040
$t = 4$	-0.208	0.019	0.044	-0.101	0.031	0.011	$c_t = 0.25$	-0.292	0.031	0.086	-0.088	0.051	0.010
$t = 5$	-0.177	0.025	0.032	-0.086	0.039	0.009	$c_t = 0.50$	-0.184	0.019	0.034	-0.091	0.022	0.009

647 C Proofs

648 C.1 Proof of Theorem 3.1

649 **Proof.** First, our simple linear regression estimator gives:

$$\hat{\tau} = (\hat{\theta}_1 + \hat{\theta}_0) - \hat{\theta}_0 = \hat{\theta}_1, \quad (31)$$

650 and

$$\hat{\theta} = (X(\mathbf{z})^\top X(\mathbf{z}))^{-1} X(\mathbf{z})^\top Y, \quad (32)$$

651 For the simplicity in the derivation, we write $X = X(\mathbf{z})$ in the following. We recap that $X(\mathbf{z}) =$
 652 $(\mathbf{1} \quad \mathbf{z})$. Since we consider complete randomization, there are n units and we assign d treatments
 653 exactly. We have:

$$X^\top X = \begin{pmatrix} n & \mathbf{z}^\top \mathbf{1} \\ \mathbf{z}^\top \mathbf{1} & \|\mathbf{z}\|_2^2 \end{pmatrix} = \begin{pmatrix} n & d \\ d & d \end{pmatrix}, \quad (33)$$

654 We remark that, for precise analysis on the bias and variance of linear regression estimator, it is
 655 critical to remove the randomness in $\det(X(\mathbf{z})^\top X(\mathbf{z}))$.

656 Naturally,

$$(X^\top X)^{-1} X^\top = \frac{1}{d(n-d)} \begin{pmatrix} d(\mathbf{1} - \mathbf{z})^\top \\ (n\mathbf{z} - d\mathbf{1})^\top \end{pmatrix}, \quad (34)$$

657 Notice that Y can be written as

$$Y = (X \quad B\mathbf{z}) \beta, \quad (35)$$

658 Here, $\beta = (\beta_0 \quad \beta_1 \quad 1)$. The shape of X is $n \times 2$, $B\mathbf{z}$ is $n \times 1$, and β is 3×1 .

659 The expression of $\hat{\theta}$ is given by

$$\begin{aligned} \hat{\theta} &= (X^\top X)^{-1} X^\top Y \\ &= (I_2 \quad (X^\top X)^{-1} X^\top B\mathbf{z}) \beta + (X^\top X)^{-1} X^\top \epsilon \\ &= \left(I_2 \quad \frac{1}{d(n-d)} \begin{pmatrix} d(\mathbf{1} - \mathbf{z})^\top B\mathbf{z} \\ (n\mathbf{z} - d\mathbf{1})^\top B\mathbf{z} \end{pmatrix} \right) \beta + (X^\top X)^{-1} X^\top \epsilon. \end{aligned} \quad (36)$$

660 Since we're only concerned with $\hat{\theta}_1$, we directly take the second term of $\hat{\theta}$:

$$\hat{\theta}_1 = \beta_1 + \frac{1}{d(n-d)} (n\mathbf{z} - d\mathbf{1})^\top B\mathbf{z} + \frac{1}{d(n-d)} (n\mathbf{z} - d\mathbf{1})^\top \epsilon, \quad (37)$$

661 Notice that $\mathbf{z} \perp \epsilon$. The **bias** is given by

$$\mathbb{E}[\hat{\theta}_1] - (\beta_1 + \frac{1}{n} \text{trace}(\mathbf{1}\mathbf{1}^\top B)) = \frac{1}{nc(1-c)} \mathbb{E}[(\mathbf{z} - c\mathbf{1})^\top B\mathbf{z}] - \frac{1}{n} \text{trace}(\mathbf{1}\mathbf{1}^\top B), \quad (38)$$

662 Next, we evaluate the term $\mathbb{E}[(\mathbf{z} - c\mathbf{1})^\top B\mathbf{z}]$, which is given by:

$$\begin{aligned} \mathbb{E}[(\mathbf{z} - c\mathbf{1})^\top B\mathbf{z}] &= \mathbb{E}[\text{trace}((\mathbf{z} - c\mathbf{1})^\top B\mathbf{z})] \\ &= \mathbb{E}[\text{trace}(\mathbf{z}(\mathbf{z} - c\mathbf{1})^\top B)] \\ &= \text{trace}(\mathbb{E}[\mathbf{z}(\mathbf{z} - c\mathbf{1})^\top] B) \\ &= \text{trace}\left(\left(c \frac{d-1}{n-1} - c^2\right) \mathbf{1}\mathbf{1}^\top B\right) \\ &= -\frac{c(1-c)}{n-1} \text{trace}(\mathbf{1}\mathbf{1}^\top B). \end{aligned} \quad (39)$$

663 Plugging it into the expression of bias, we get:

$$\text{Bias}(\hat{\tau}_{(1)}) = -\frac{\sum_{i,j} B_{ij}}{n} \left(1 + \frac{1}{n-1}\right) = -\frac{\sum_{i,j} B_{ij}}{n-1}. \quad (40)$$

Subsequently, we derive the expression of variance. Noting that $\mathbf{z} \perp \epsilon$ and $\mathbb{E}[\epsilon] = \mathbf{0}$, we have:

$$\text{Cov}[(n\mathbf{z} - d\mathbf{1})^\top B\mathbf{z}, (n\mathbf{z} - d\mathbf{1})^\top \epsilon] = 0, \quad (41)$$

Thus, we can omit all the covariance terms in the expression of variance, yielding:

$$\text{Var}(\hat{\tau}) = \left(\frac{1}{nc(1-c)}\right)^2 \text{Var}((\mathbf{z} - c\mathbf{1})^\top B\mathbf{z}) + \frac{1}{(nc(1-c))^2} \mathbb{E}[(\mathbf{z} - c\mathbf{1})^\top (\mathbf{z} - c\mathbf{1})] \sigma_e^2, \quad (42)$$

First, we consider the second term above, which represents the variance induced by ϵ :

$$\begin{aligned} \text{Var}_e &= \frac{1}{(d(n-d))^2} \mathbb{E}[(n\mathbf{z} - d\mathbf{1})^\top (n\mathbf{z} - d\mathbf{1})] \sigma_e^2 \\ &= \left(\frac{n}{d(n-d)}\right)^2 \mathbb{E}[(\mathbf{z} - \frac{d}{n}\mathbf{1})^\top (\mathbf{z} - \frac{d}{n}\mathbf{1})] \sigma_e^2 \\ &= \left(\frac{n\sigma_e}{d(n-d)}\right)^2 \mathbb{E}[d - \frac{2d}{n} \times d + \frac{d^2}{n}] \\ &= \left(\frac{n\sigma_e}{d(n-d)}\right)^2 (d - \frac{d^2}{n}) \\ &= \frac{\sigma_e^2}{nc(1-c)}. \end{aligned} \quad (43)$$

Next, we discuss the first term:

$$\left(\frac{1}{nc(1-c)}\right)^2 \text{Var}((\mathbf{z} - c\mathbf{1})^\top B\mathbf{z}), \quad (44)$$

We use the expansion $\text{Var}(X) = \mathbb{E}[X^2] - E[X]^2$, and the square of expectation term has been simplified in equation (39). Thus, our focus is the expectation of square term, which can be expanded with brute-force:

$$\left(\frac{1}{c(1-c)n}\right)^2 \sum_{i,j,k,l} B_{ij} B_{kl} \mathbb{E}[(z_i - c)(z_k - c)z_j z_l], \quad (45)$$

As mentioned in the main paper, our technique involves classifying the index set $I \subset [n]^4$ that restricts the quadruples (i, j, k, l) and calculating $\mathbb{E}[(z_i - c)(z_k - c)z_j z_l]$ respectively. Since $B_{ii} = 0$ for all $i \in [n]$, we do not need to consider the cases where $i = j$ or $k = l$. We present the result as follows:

1. $i = k, j = l, i \neq k$,

$$\begin{aligned} \mathbb{E}[(z_i - c)(z_k - c)z_j z_l] &= \mathbb{E}[(z_i - c)^2 z_j^2] \\ &= \mathbb{E}[(z_i - 2cz_i + c^2)z_j] \\ &= (1 - 2c)\mathbb{E}[z_i z_j] + c^2 \mathbb{E}[z_j] \\ &= (1 - 2c)c \frac{d-1}{n-1} + c^3 \\ &= c^2(1-c) + O\left(\frac{1}{n}\right). \end{aligned} \quad (46)$$

2. $i = l, k = j, i \neq k$

$$\begin{aligned} \mathbb{E}[(z_i - c)(z_k - c)z_j z_l] &= \mathbb{E}[(z_i - c)(z_k - c)z_i z_k] \\ &= \mathbb{E}[(1-c)^2 z_i z_k] \\ &= (1-c)^2 c \frac{d-1}{n-1} \\ &= (1-c)^2 c^2 + O\left(\frac{1}{n}\right). \end{aligned} \quad (47)$$

677 3. Only $i = k$ holds, with i, j , and l being distinct.

$$\begin{aligned}
\mathbb{E}[(z_i - c)(z_k - c)z_j z_l] &= \mathbb{E}[(z_i - c)^2 z_k z_l] \\
&= (1 - 2c)\mathbb{E}[z_i z_k z_l] + c^2 \mathbb{E}[z_k z_l] \\
&= (1 - 2c)c \frac{(d-1)(d-2)}{(n-1)(n-2)} + c^3 \frac{d-1}{n-1} \\
&= c^3(1 - c) + O\left(\frac{1}{n}\right).
\end{aligned} \tag{48}$$

678 4. Only $j = l$ holds, with i, j , and k being distinct.

$$\begin{aligned}
\mathbb{E}[(z_i - c)(z_k - c)z_j z_l] &= \mathbb{E}[(z_i - c)(z_k - c)z_j] \\
&= \frac{nd(d-n) + 2d^2 + 2n^2 - 4nd}{n^2(n-1)(n-2)} \\
&= \frac{c(d-n) + 2c^2 + 2 - 4c}{(n-1)(n-2)} \\
&= \frac{c(c-1)}{n} + O\left(\frac{1}{n^2}\right).
\end{aligned} \tag{49}$$

679 5. i, j, k, l are all distinct.

$$\begin{aligned}
&\mathbb{E}[(z_i - c)(z_k - c)z_j z_l] \\
&= \mathbb{E}[(z_i z_k - c(z_i + z_k) + c^2)z_j z_l] \\
&= \frac{d(d-1)(d-2)(d-3)}{n(n-1)(n-2)(n-3)} - 2c \frac{d(d-1)(d-2)}{n(n-1)(n-2)} + c^2 \frac{d(d-1)}{n(n-1)} \\
&= \frac{d(d-1)}{n(n-1)} \left(\frac{(d-2)(d-3)}{(n-2)(n-3)} + \frac{d^2}{n^2} - 2 \frac{d(d-2)}{n(n-2)} \right) \\
&= \frac{d(d-1)}{n(n-1)} \left(\frac{(d^2 - 5d + 6)n^2 + d^2(n^2 - 5d + 6) - 2dn(dn - 2n - 3d + 6)}{n^2(n-2)(n-3)} \right) \\
&= \frac{d(d-1)}{n(n-1)} \left(\frac{(d^2 - 5d + 6)n^2 + d^2(n^2 - 5d + 6) - 2dn(dn - 2n - 3d + 6)}{n^2(n-2)(n-3)} \right) \\
&= \frac{c(d-1)}{(n-1)} \left(\frac{c(d-n) + 6 + 6c^2 - 12c}{(n-2)(n-3)} \right) \\
&= \frac{c^3(c-1)}{n} + O\left(\frac{1}{n^2}\right).
\end{aligned} \tag{50}$$

680 Combine these five cases and corresponding index sets, we get the expression of variance as presented
681 in Theorem 1. For the convenience of proof of Corollary 3.3 later, we also annotate the order of each
682 term at last line. $\square$

683 C.2 Proof of Corollary 3.3

684 **Proof.** Since Assumption 3.2 restricts

$$\sum_{i,j} B_{ij} = \Theta(n), \tag{51}$$

685 We get the order of bias instantly:

$$\text{Bias}(\hat{\tau}_{(1)}) = -\frac{\sum_{i,j} B_{ij}}{n-1} = \Theta(1). \tag{52}$$

686 Next, we discuss the order of variance. It concerns bounding the order of the summations over five
687 index sets mentioned in Appendix C.1. To begin with, we notice that the variance is non-negative and
688 the variance induced by ϵ is:

$$\frac{\sigma_e^2}{nc(1-c)} = \Theta\left(\frac{1}{n}\right), \tag{53}$$

Thus, we can first claim that

$$\text{Var}(\hat{\tau}) = \Omega\left(\frac{1}{n}\right), \quad (54)$$

We then prove the first term in the expression of variance has a order of $O(1/n)$. Specifically, we bound those five summations respectively. According to the Assumption 3.2, we get a naive bound that will be used repeatedly:

$$|B_{ij}| \leq \sum_j |B_{ij}| \leq b. \quad (55)$$

1. $i = k, j = l, i \neq k$.

$$\sum_{ij} B_{ij}^2 \leq \sum_i b \sum_j |B_{ij}| \leq b^2 n, \quad (56)$$

Thus,

$$\sum_{ij} B_{ij}^2 = O(n). \quad (57)$$

2. $i = l, k = j, i \neq k$

$$|\sum_{i,j} B_{ij} B_{ji}| \leq \sum_{i,j} |B_{ij} B_{ji}| \leq b \sum_{i,j} B_{ij} \leq b^2 n, \quad (58)$$

Thus,

$$\sum_{ij} B_{ij} B_{ji} = O(n). \quad (59)$$

3. Only $i = k$ holds, with i, j , and l being distinct.

$$|\sum_{i,j,l} B_{ij} B_{il}| \leq \sum_{i,j,l} |B_{ij} B_{il}| = \sum_{i,j} |B_{ij}| \sum_l |B_{il}| \leq b \sum_{i,j} |B_{ij}| \leq b^2 n, \quad (60)$$

Thus,

$$\sum_{i,j,l} B_{ij} B_{il} = O(n). \quad (61)$$

4. Only $j = l$ holds, with i, j , and k being distinct.

$$|\sum_{i,j,k} B_{ij} B_{kj}| \leq \sum_{i,j,k} |B_{ij} B_{kj}| \leq b \sum_{i,j,k} |B_{ij}| \leq b^2 n^2, \quad (62)$$

Thus,

$$\sum_{i,j,k} B_{ij} B_{kj} = O(n^2). \quad (63)$$

5. i, j, k, l are all distinct.

$$\sum_{i,j,k,l} B_{ij} B_{kl} = \left(\sum_{ij} B_{ij}\right)^2 = \Theta(n^2). \quad (64)$$

Combine the orders of five summations above and the order of corresponding $\mathbb{E}[(z_i - c)(z_k - c)z_j z_l]$, we get:

$$\text{Var}(\hat{\tau}) = \Theta\left(\frac{1}{n}\right), \quad (65)$$

Finally, we comment that the bound on $|\sum_{i,j,k} B_{ij} B_{kj}|$ is rough, while we do not apply the third part of Assumption 3.2, i.e. restricting the order $\sum_{i,j,k} B_{ij} B_{kj} = \sum_j (\sum_i B_{ij})^2$ directly.

Such a rough bound is still acceptable because of the equation (49), i.e.

$$\mathbb{E}[(z_i - c)(z_k - c)z_j z_l] = \Theta\left(\frac{1}{n}\right), \quad (66)$$

For the situation of merging $T \geq 2$ experimental data, if $\frac{\sum_{t=1}^T c_t}{T} \neq \mathbb{E}[z_i]$, we would have:

$$\mathbb{E}\left[\left(z_i - \frac{\sum_{t=1}^T c_t}{T}\right)\left(z_k - \frac{\sum_{t=1}^T c_t}{T}\right)z_j z_l\right] = \Theta(1), \quad (67)$$

This explains why we need a further assumption to control the order of variance there. $\square$

709 **C.3 Proof of Theorem 3.5**

710 **Proof.** In the situation of $T = 2$, the sample size increases from n to $2n$, and the number of treated
711 units changes from d to $d_1 + d_2$. Correspondingly, the treatment proportions in two experiments are
712 c_1 and c_2 . In linear regression, this additionally leads to the following two changes:

$$B \rightarrow \begin{pmatrix} B & 0 \\ 0 & B \end{pmatrix} \quad \mathbf{z} \rightarrow \begin{pmatrix} \mathbf{z}_1 \\ \mathbf{z}_2 \end{pmatrix}. \quad (68)$$

713 We make the changes above and get

$$\begin{aligned} & \text{Bias}(\hat{\tau}_{(2)}) / \left(\sum_{i,j} B_{ij} \right) + 1 \\ &= \frac{2n}{(d_1 + d_2)(2n - (d_1 + d_2))} \text{trace} \left(\mathbb{E} \left[\mathbf{z} \left(\mathbf{z} - \frac{d_1 + d_2}{2n} \mathbf{1} \right)^\top \right] \begin{pmatrix} B & 0 \\ 0 & B \end{pmatrix} \right) \\ &= \frac{2n}{(d_1 + d_2)(2n - (d_1 + d_2))} \text{trace} \left(\left(\mathbb{E} [\mathbf{z} \mathbf{z}^\top] - \frac{d_1 + d_2}{2n} \mathbb{E} [\mathbf{z} \mathbf{1}^\top] \right) \begin{pmatrix} B & 0 \\ 0 & B \end{pmatrix} \right) \\ &= \frac{2n}{(d_1 + d_2)(2n - (d_1 + d_2))} \left(\text{trace} ((\mathbb{E} [\mathbf{z}_1 \mathbf{z}_1^\top] + \mathbb{E} [\mathbf{z}_2 \mathbf{z}_2^\top]) B) - \frac{d_1 + d_2}{2n} \left(\frac{d_1}{n} + \frac{d_2}{n} \right) \text{trace} (\mathbf{1} \mathbf{1}^\top B) \right) \\ &= \frac{2n}{(d_1 + d_2)(2n - (d_1 + d_2))} \left(\frac{d_1(d_1 - 1) + d_2(d_2 - 1)}{n(n - 1)} - \frac{(d_1 + d_2)^2}{2n^2} \right) \text{trace} (\mathbf{1} \mathbf{1}^\top B) \\ &= \frac{2n^2}{(d_1 + d_2)(2n - (d_1 + d_2))} \frac{(n - 1)(d_1 - d_2)^2 + 2(d_1^2 + d_2^2) - 2n(d_1 + d_2)}{2n^2(n - 1)} \\ &= \frac{(n - 1)(d_1 - d_2)^2 + 2(d_1^2 + d_2^2) - 2n(d_1 + d_2)}{(n - 1)(d_1 + d_2)(2n - (d_1 + d_2))} \\ &= \frac{(n - 1)(c_1 - c_2)^2 + 2(c_1^2 + c_2^2) - 2(c_1 + c_2)}{(n - 1)(c_1 + c_2)(2 - (c_1 + c_2))}. \end{aligned} \quad (69)$$

714 Thus, we get:

$$\text{Bias}(\hat{\tau}_{(2)}) = - \frac{\sum_{i,j} B_{ij}}{n} \left(1 - \frac{(n - 1)(c_1 - c_2)^2 + 2(c_1^2 + c_2^2) - 2(c_1 + c_2)}{(n - 1)(c_1 + c_2)(2 - (c_1 + c_2))} \right). \quad (70)$$

715 **C.4 Proof of Theorem 3.6**

716 **Proof.** As mentioned in the main paper, there is no additional challenge in deriving the result of
717 $T > 2$ case. Refer to the fourth equation in equation (69), we now have:

$$\begin{aligned} \text{Bias}(\hat{\tau}_{(T)}) / \left(\sum_{i,j} B_{ij} \right) + 1 &= \frac{Tn}{\sum_{t=1}^T d_t (Tn - \sum_{t=1}^T d_t)} \left(\frac{\sum_{t=1}^T d_t (d_t - 1)}{n(n - 1)} - \frac{(\sum_{t=1}^T d_t)^2}{Tn^2} \right) \\ &= \frac{T}{n(\sum_{t=1}^T c_t)(T - \sum_{t=1}^T c_t)} \left(\frac{T \sum_{t=1}^T c_t^2 - (\sum_{t=1}^T c_t)^2}{T} + O\left(\frac{1}{n}\right) \right). \end{aligned} \quad (71)$$

718 Thus, since $\sum_{i,j} B_{ij} = \Theta(n)$, we have:

$$\text{Bias}(\hat{\tau}_{(T)}) = - \frac{\sum_{i,j} B_{ij}}{n} \left(1 - \frac{T \sum_{t=1}^T c_t^2 - (\sum_{t=1}^T c_t)^2}{(\sum_{t=1}^T c_t)(T - \sum_{t=1}^T c_t)} \right) + O\left(\frac{1}{n}\right). \quad (72)$$

719 **C.5 Proof of Corollary A.1**

720 **Proof.** We continue to use the notation in the main paper, and further bias reduction corresponds to:

$$\frac{kC_2 - C_1^2}{C_1(k - C_1)} < \frac{(k + 1)(C_2 + x^2) - (C_1 + x)^2}{(C_1 + x)(k + 1 - C_1 - x)}, \quad (73)$$

721 We expand the two sides of this inequality with brute-force, and get following condition:

$$\begin{aligned}
& k(k+1)C_1C_2 + k(k+1)C_1x^2 - (k+1)C_1^2C_2 - (k+1)C_1^2x^2 \\
& - kC_1^3 - 2kC_1^2x - kC_1x^2 + C_1^4 + 2C_1^3x + C_1^2x^2 \\
& > k(k+1)C_1C_2 + k(k+1)C_2x - kC_1^2C_2 - 2kC_1C_2x \\
& - kC_2x^2 - (k+1)C_1^3 + C_1^4 + C_1^3x - (k+1)C_1^2x + C_1^3x + C_1^2x^2,
\end{aligned} \tag{74}$$

722 which is equivalent to:

$$(k^2C_1 - (k+1)C_1^2 + kC_2)x^2 - ((k-1)C_1^2 + 2kC_1C_2 + k(k+1)C_2)x + C_1^3 - C_1^2C_2 > 0. \tag{75}$$

723 $\square$

724 C.6 Lemma C.1 for Variance Analysis with Proof

725 We construct the following lemma for the convenience of variance analysis:

726 **Lemma C.1.** *Given Assumption 3.2, for a treatment vector $\mathbf{z}$ with treatment proportion $c \in (0, 1)$*
727 *and any constant $w \in (0, 1)$, we have:*

$$\text{Var}((\mathbf{z} - w\mathbf{1})^\top B\mathbf{z}) = O\left(\frac{1}{n}\right). \tag{76}$$

728 **Proof.** First, we will continue to apply the idea of expansion in the proof of variance part of Theorem 1.
729 However, the proof in the Theorem 1 can't apply directly because the order of $\mathbb{E}[(z_i - c)(z_k - c)z_j z_l]$
730 increases from $\Theta(1/n)$ to $\Theta(1)$, when $w \neq c$. These two cases are:

- 731 1. Only $j = l$ holds, with i, j , and k being distinct.
- 732 2. i, j, k, l are all distinct.

733 For the first case, we resort to the third part of Assumption 3.2:

$$\sum_{i,j,k} B_{ij}B_{kj} = \sum_j \left(\sum_i B_{ij}\right)^2 = O(n), \tag{77}$$

734 For the second case, we will prove that the two coefficients of term $(\sum_{ij} B_{ij})^2$ in the expression of
735 $\text{Var}((\mathbf{z} - w\mathbf{1})^\top B\mathbf{z})$ can cancel out. Namely, we will prove:

$$\mathbb{E}[(z_i - w)(z_k - w)z_j z_l] - (\mathbb{E}[(\mathbf{z} - w\mathbf{1})^\top B\mathbf{z}] / \text{trace}(\mathbf{1}\mathbf{1}^\top B))^2 = O\left(\frac{1}{n}\right), \tag{78}$$

736 for the case i, j, k, l are not equal to each other.

737 Notice that, for $s = 1, 2, 3$, we have:

$$\frac{d-s}{n-s} = c + O\left(\frac{1}{n}\right), \tag{79}$$

738 On one hand,

$$\begin{aligned}
\mathbb{E}[(z_i - w)(z_k - w)z_j z_l] &= \mathbb{E}[(z_i - w)(z_k - w) \mid z_j = z_l = 1] \left(c^2 + O\left(\frac{1}{n}\right)\right) \\
&= (c - w)^2 c^2 + O\left(\frac{1}{n}\right).
\end{aligned} \tag{80}$$

739 On the other hand,

$$\begin{aligned}
\frac{\mathbb{E}[(\mathbf{z} - w\mathbf{1})^\top B\mathbf{z}]}{\text{trace}(\mathbf{1}\mathbf{1}^\top B)} &= \frac{\text{trace}(\mathbb{E}[\mathbf{z}(\mathbf{z} - w\mathbf{1})^\top] B)}{\text{trace}(\mathbf{1}\mathbf{1}^\top B)} \\
&= \frac{\text{trace}((\mathbb{E}[\mathbf{z}\mathbf{z}^\top] - cw\mathbf{1}\mathbf{1}^\top)B)}{\text{trace}(\mathbf{1}\mathbf{1}^\top B)} \\
&= c(c - w) + O\left(\frac{1}{n}\right).
\end{aligned} \tag{81}$$

740 In the last equation, we again utilize $B_{ii} = 0$ for all unit $i \in [n]$. Combine two equations above,
 741 we successfully control the order of coefficient of term $(\sum_{ij} B_{ij})^2$, in the expression of concerned
 742 variance $\text{Var}((\mathbf{z} - w\mathbf{1})^\top B\mathbf{z})$.

743 For other three cases of $\mathbb{E}[(z_i - c)(z_k - c)z_j z_l]$, the proof of Theorem 1 can be utilized directly here,
 744 because $\mathbb{E}[(z_i - c)(z_k - c)z_j z_l] = \Theta(1)$ there, and the order of this term can't exceed constant no
 745 matter what value w takes. Therefore, we finish the proof. $\square$

746 C.7 Proof of Theorem 3.7

747 **Proof.** We first present the proof for the case of **temporally independent** T experiments is a direct
 748 extension of Lemma C.1. In fact, with Lemma C.1, we can conclude that

$$\text{Var} \left(\sum_{t=1}^T \left(\mathbf{z}_t - \frac{\sum_{k=1}^T c_k}{T} \mathbf{1} \right)^\top B \mathbf{z}_t \right) = \sum_{t=1}^T \text{Var} \left(\left(\mathbf{z}_t - \frac{\sum_{k=1}^T c_k}{T} \mathbf{1} \right)^\top B \mathbf{z}_t \right) = O \left(\frac{1}{n} \right), \quad (82)$$

749 On the other hand, the order of variance introduced by ϵ is always $\Theta(1/n)$, and the concerned
 750 variance is the sum of these non-negative variance terms. Therefore, we can conclude that

$$\text{Var}(\hat{\tau}) = \Theta \left(\frac{1}{n} \right), \quad (83)$$

751 Next, we study the case of **staggered rollout** experiments. The complexity here is strong temporal
 752 covariance of treatments.

753 Based on the Lemma C.1, the remaining work is bounding the sum of covariance terms. We have:

$$\begin{aligned} \text{Var} \left(\sum_{t=1}^T \left(\mathbf{z}_t - \frac{\sum_{k=1}^T c_k}{T} \mathbf{1} \right)^\top B \mathbf{z}_t \right) &= \sum_{t=1}^T \text{Var} \left(\left(\mathbf{z}_t - \frac{\sum_{k=1}^T c_k}{T} \mathbf{1} \right)^\top B \mathbf{z}_t \right) \\ &+ 2 \sum_{t_1 < t_2} \text{Cov} \left[\left(\mathbf{z}_{t_1} - \frac{\sum_{k=1}^T c_k}{T} \mathbf{1} \right)^\top B \mathbf{z}_{t_1}, \left(\mathbf{z}_{t_2} - \frac{\sum_{k=1}^T c_k}{T} \mathbf{1} \right)^\top B \mathbf{z}_{t_2} \right]. \end{aligned} \quad (84)$$

754 For simplicity, we use $\mathbf{u}, \mathbf{v}$ to denote $\mathbf{z}_{t_1}, \mathbf{z}_{t_2}$, respectively, and use $w = \sum_{t=1}^T c_t / T$. We continue to
 755 expand the covariance term, and get:

$$\sum_{i,j,k,l} B_{ij} B_{kl} (\mathbb{E}[(u_i - w) u_j (v_k - w) v_l] - \mathbb{E}[(u_i - w) u_j] \mathbb{E}[(v_k - w) v_l]), \quad (85)$$

756 Correspondingly, we need to discuss the order of the this term in five cases, as well as that in the
 757 Appendix C.1. Similarly, for the first four cases (except for the case where i, j, k, l are not equal to
 758 each other), we allow for:

$$\mathbb{E}[(u_i - w) u_j (v_k - w) v_l] - \mathbb{E}[(u_i - w) u_j] \mathbb{E}[(v_k - w) v_l] = \Theta(1), \quad (86)$$

759 Therefore, we only need to prove:

$$\mathbb{E}[(u_i - w) u_j (v_k - w) v_l] - \mathbb{E}[(u_i - w) u_j] \mathbb{E}[(v_k - w) v_l] = O \left(\frac{1}{n} \right), \quad (87)$$

760 when i, j, k, l are not equal to each other.

761 Since $\mathbf{u}$ is the treatment vector that is assigned before $\mathbf{v}$, we should first condition on u_i, u_j .

$$\begin{aligned} \mathbb{E}[(u_i - w) u_j (v_k - w) v_l] &= \left(c_{t_1}^2 + O \left(\frac{1}{n} \right) \right) (1 - w) \mathbb{E}[(v_k - w) v_l \mid u_i = u_j = 1] \\ &\quad - \left(c_{t_1} (1 - c_{t_1}) + O \left(\frac{1}{n} \right) \right) w \mathbb{E}[(v_k - w) v_l \mid u_j = 1] \\ &= (1 - w) c_{t_1}^2 (c_{t_2} - w) c_{t_2} - w c_{t_1} (1 - c_{t_1}) (c_{t_2} - w) c_{t_2} + O \left(\frac{1}{n} \right) \\ &= (c_{t_1} - w) c_{t_1} (c_{t_2} - w) c_{t_2} + O \left(\frac{1}{n} \right). \end{aligned} \quad (88)$$

762 On the other hand, it is easy to check that

$$\mathbb{E}[(u_i - w)u_j] \mathbb{E}[(v_k - w)v_l] = (c_{t_1} - w)c_{t_1}(c_{t_2} - w)c_{t_2} + O\left(\frac{1}{n}\right), \quad (89)$$

763 Thus, we finish the proof. $\square$

764 *Remark C.2.* In fact, the covariance term in other four cases are likely of constant order. For example,
765 we check the case where only $j = l$ holds:

$$\begin{aligned} \mathbb{E}[(u_i - w)u_j(v_k - w)v_l] &= \mathbb{E}[(u_i - w)u_j(v_k - w)v_j] \\ &= c_{t_1} \mathbb{E}[(u_i - w)(v_k - w)v_j \mid u_j = 1] \\ &= c_{t_1} \left(c_{t_1} - w + O\left(\frac{1}{n}\right) \right) \left(c_{t_2} - w + O\left(\frac{1}{n}\right) \right) \\ &= c_{t_1}(c_{t_1} - w)(c_{t_2} - w) + O\left(\frac{1}{n}\right). \end{aligned} \quad (90)$$

766 Therefore, in this case, the value of covariance term

$$\begin{aligned} &\mathbb{E}[(u_i - w)u_j(v_k - w)v_l] - \mathbb{E}[(u_i - w)u_j] \mathbb{E}[(v_k - w)v_l] \\ &= c_{t_1}(c_{t_1} - w)(c_{t_2} - w) - c_{t_1}c_{t_2}(c_{t_1} - w)(c_{t_2} - w) + O\left(\frac{1}{n}\right) \\ &= \Theta(1), \end{aligned} \quad (91)$$

767 if $c_{t_1} > 0$ and $c_{t_2} < 1$. Therefore, the third part of Assumption 3.2 is still necessary here.

NeurIPS Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [Yes]

Justification: The abstract and introduction state the main contributions, assumptions, and scope, and these are supported by the theory and experiments in Sections 1–5 and the appendix.

Guidelines:

- The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
- The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
- The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
- It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification: The paper discusses key limitations and scope restrictions, including the focus on static GATE, suppression of temporal interference, and the simplifying assumptions behind the exact theory; see Sections 3–6 and Appendix A.

Guidelines:

- The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
- The authors are encouraged to create a separate “Limitations” section in their paper.
- The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
- The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
- The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
- The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
- If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
- While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren’t acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [Yes]

Justification: The theoretical results are stated with explicit assumptions and cross-referenced proofs, with detailed derivations deferred to the appendix and supplemental material.

Guidelines:

- The answer [N/A] means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes]

Justification: The paper specifies the experimental design, datasets, models, and evaluation protocol, and we will further ensure reproducibility by providing runnable code in the supplementary material.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general, releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
 - (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
 - (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
 - (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
 - (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in

some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [Yes]

Justification: We will provide anonymized code and instructions in the supplementary material so that the main experimental results can be faithfully reproduced.

Guidelines:

- The answer [N/A] means that paper does not include experiments requiring code.
- Please see the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
- The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. Experimental setting/details

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [Yes]

Justification: The paper specifies the network data, interference models, merging schemes, randomization schemes, and main model choices in Section 5 and Appendix B, and the supplementary code will include full training details.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
- The full details can be provided either with the code, in appendix, or as supplemental material.

7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes]

Justification: The experiments report variability information through standard deviations, and the paper explicitly states that the bias and standard deviation are computed from Monte Carlo repetitions; see Section 5.2 and the result tables.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.

- The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
- If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. Experiments compute resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [No]

Justification: Computational cost is not a central concern of this paper, whose main focus is the statistical properties of regression-based estimators under network interference. The experiments use lightweight models and standard simulation pipelines, and we will provide implementation details in the supplementary material.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
- The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
- The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. Code of ethics

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics <https://neurips.cc/public/EthicsGuidelines>?

Answer: [Yes]

Justification: To the best of our knowledge, the research complies with the NeurIPS Code of Ethics and does not involve unethical data collection or human-subject procedures.

Guidelines:

- The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.
- If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
- The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. Broader impacts

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [No]

Justification: The current submission does not include a dedicated discussion of broader societal impacts. This is a methodological paper on experimental design and estimation.

Guidelines:

- The answer [N/A] means that there is no societal impact of the work performed.
- If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
- Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. Safeguards

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [N/A]

Justification: The paper does not release high-risk models or scraped datasets, so special safeguards for responsible release are not applicable here.

Guidelines:

- The answer [N/A] means that the paper poses no such risks.
- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
- Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
- We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. Licenses for existing assets

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [No]

Justification: Existing assets such as the FB-Stanford3 network and software packages are credited by citation, but the manuscript does not yet explicitly document licenses and terms of use. We will add this information in the supplementary material.

Guidelines:

- The answer [N/A] means that the paper does not use existing assets.
- The authors should cite the original paper that produced the code package or dataset.
- The authors should state which version of the asset is used and, if possible, include a URL.
- The name of the license (e.g., CC-BY 4.0) should be included for each asset.
- For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
- If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, paperswithcode.com/datasets has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
- For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
- If this information is not available online, the authors are encouraged to reach out to the asset’s creators.

13. New assets

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [Yes]

Justification: We will release code for the proposed method and experiments, together with documentation and instructions in the supplementary material.

Guidelines:

- The answer [N/A] means that the paper does not release new assets.
- Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
- The paper should discuss whether and how consent was obtained from people whose asset is used.
- At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. Crowdsourcing and research with human subjects

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [N/A]

Justification: The paper does not involve crowdsourcing or experiments with human subjects.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
- According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. Institutional review board (IRB) approvals or equivalent for research with human subjects

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

Answer: [N/A]

1083 Justification: The paper does not involve human subjects, so IRB approval or an equivalent
1084 review is not applicable.

1085 Guidelines:

- 1086 • The answer [N/A] means that the paper does not involve crowdsourcing nor research
1087 with human subjects.
- 1088 • Depending on the country in which research is conducted, IRB approval (or equivalent)
1089 may be required for any human subjects research. If you obtained IRB approval, you
1090 should clearly state this in the paper.
- 1091 • We recognize that the procedures for this may vary significantly between institutions
1092 and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the
1093 guidelines for their institution.
- 1094 • For initial submissions, do not include any information that would break anonymity (if
1095 applicable), such as the institution conducting the review.

1096 16. Declaration of LLM usage

1097 Question: Does the paper describe the usage of LLMs if it is an important, original, or
1098 non-standard component of the core methods in this research? Note that if the LLM is used
1099 only for writing, editing, or formatting purposes and does *not* impact the core methodology,
1100 scientific rigor, or originality of the research, declaration is not required.

1101 Answer: [N/A]

1102 Justification: LLMs are not an important or non-standard component of the core methodology
1103 in this research; any LLM use was limited to writing assistance only, if any.

1104 Guidelines:

- 1105 • The answer [N/A] means that the core method development in this research does not
1106 involve LLMs as any important, original, or non-standard components.
- 1107 • Please refer to our LLM policy in the NeurIPS handbook for what should or should not
1108 be described.